

Glaucoma — Case Cards

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Every card follows the same headings. The generic long-case proforma is in 00 · Index and master case format.

★ kept at JEH last year · Long case · Short case · Task · Chart

Contents

No.	Card	Type	Last year
G1	Primary open-angle glaucoma, including the glaucomatous optic disc	Long + short case	★ Kept 6 times (one with pseudoexfoliation glaucoma)
G4	Primary angle-closure disease — PACS, PAC and PACG, including after laser iridotomy	Long case	★ Kept 4 times (one as the long case)
G5	Neovascular glaucoma, including rubeosis iridis	Long case	★ Kept once
G6	Pseudoexfoliation syndrome and pseudoexfoliation glaucoma	Long + short case	★ Kept once (with POAG)
G7	CRVO with POAG, including hemi-CRVO with POAG	Short case	★ Kept twice (CRVO with POAG as a glaucoma case; hemi-CRVO with POAG as a fundus case)
G8	Applanation tonometry — perform and record	Task	★ Done in 8 of the 12 glaucoma cases
G9	Gonioscopy — perform, grade and draw	Task	★ Done in about 8 of the 12 glaucoma cases
G10	Humphrey visual field in glaucoma — reading the printout	Chart	★ Done in 3 of the 12 glaucoma cases
G2	POAG with trabeculectomy — bleb assessment and the failed bleb	Long case	—
G3	POAG with a glaucoma drainage device	Long case	—

G1 Primary open-angle glaucoma, including the glaucomatous optic disc

★ Kept 6 times at JEH last year (one with pseudoexfoliation glaucoma) · Long case and short case

At a glance

- **Primary open-angle glaucoma (POAG):** a chronic, progressive optic neuropathy with an **open, normal-looking angle** and reduced outflow; usually asymptomatic, found on routine examination or follow-up.
- Commonly kept: stable signs (cupped disc, measurable pressure, open angle). The **glaucomatous disc** is also a short case.
- Examiners test the **disc description in order**, intraocular pressure (IOP) with method, time and central corneal thickness (CCT), gonioscopy, target pressure and drugs.

Case format — history

Chief complaint: “Mr X, a 70-year-old man, has come for regular follow-up for glaucoma” — or “...came with complaints of defective vision in both eyes for 1 year”.

House wording: “Patient was apparently normal ___ years ago, then had defective vision in both eyes, gradual in onset, progressive in nature, not associated with pain or headache. He was found to have increased pressure elsewhere and started on eye drops (1 drop at bedtime).”

What to ask	Why it matters
History of present illness: onset, progression; blurring at one time of day	Gradual, often silent until late; blurring matches the IOP peak
Frequent change of near glasses	Pressure on zonules and ciliary body impairs accommodation; field loss mistaken for poor vision
Night blindness	Defective dark adaptation; worse on miotics
Old records: baseline IOP, fields, optical coherence tomography (OCT)	Target pressure; progression
Negative history: no headache with vomiting, coloured haloes, redness, watering or pain	Rules out angle closure (haloes = corneal oedema in a pressure rise)
No trauma; no steroids (drops, nasal spray, tablets)	Rules out angle-recession and steroid-induced glaucoma
No sudden loss of vision	Rules out central retinal vein occlusion , an acute IOP rise, snuff-out after trabeculectomy
Past, drug and surgical: diabetes, hypertension, thyroid disease, migraine, Raynaud's, peripheral vascular disease, stroke, head injury, blood loss	Risk factors; vasospasm matters most in normal-tension glaucoma (NTG); shock or head injury can mimic it
Asthma, chronic obstructive pulmonary disease (COPD), heart failure, heart block; sulpha or other drug allergy, kidney disease	No beta-blocker; no carbonic anhydrase inhibitor (CAI)
Eye surgery or laser: cataract, trabeculectomy, refractive, peripheral iridotomy (PI)	Status (pseudophakia, bleb); refractive surgery thins the cornea: IOP reads low
Treatment history: “E/D (eye drops) timolol 0.5% BD (twice daily)”	Drug, strength, frequency; drops for > 3 years mean mitomycin C at trabeculectomy
Personal and family: glaucoma in relatives	Risk 4-fold in siblings, 2-fold in offspring

Case format — examination

General and systemic (this case only)

- **Pulse, blood pressure, chest:** bradycardia, heart block or wheeze bars beta-blockers; low diastolic perfusion pressure is a risk factor.
- **Face and skin:** haemangioma (Sturge-Weber syndrome), neurofibromatosis, carotid-cavernous fistula: raised episcleral venous pressure.

Ocular examination — RE | LE template

Right eye (RE) and left eye (LE) columns.

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Central vision is lost last	6/18, pinhole 6/12
Extraocular movements (EOM)	Seniors' addition	Full

Structure	Look for in this case	How to record (example)
Intraocular pressure	Goldmann applanation tonometer (GAT) first; asymmetry ≥ 4 mm Hg is significant	Tn (tension) by GAT 18 mm Hg at 10 am
Lids	Prostaglandin (PG) analogue: pigmentation, long lashes	Normal
Conjunctiva	Drug congestion, worse inferiorly; bleb	Clear
Cornea	Krukenberg spindle; pseudoexfoliation (PXF)	Clear
Anterior chamber	Van Herick (VH) grade; cells, flare	VH grade IV (ND, normal depth)
Iris	PXF; transillumination defects; PI	CPN (colour, pattern normal)
Pupil	Relative afferent pupillary defect (RAPD) = advanced	RTL 3 mm (round, reacting to light)
Lens	PXF on the capsule; glaucomflecken	PCIOL (posterior chamber intraocular lens)
Gonioscopy	Open angle; no peripheral anterior synechiae (PAS), recession or new vessels	Scleral spur (SS) seen in all quadrants
Fundus	Disc in slide order; other causes of field loss (pallor, tilted disc, drusen, pit)	Disc vertically oval, cup-disc ratio (CDR) 0.8, neuroretinal rim (NRR) thinning + (superior, inferior), nasalisation +, bayoneting +, peripapillary atrophy (PPA) +
Fields	Confrontation (seniors' addition)	Full in all quadrants

Special tests

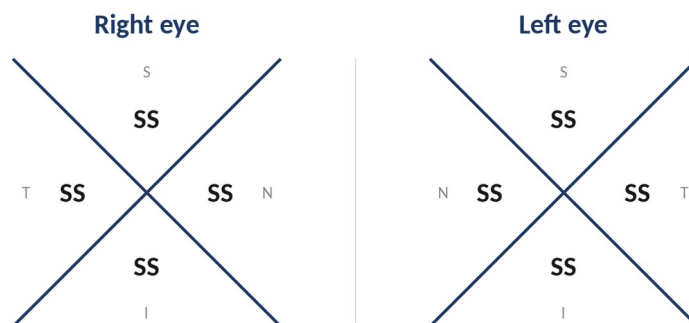
Applanation tonometry (card G8)

1. Do it first: gonioscopy lowers the reading; dilatation raises it by 4–5 mm Hg.
2. GAT in each eye; note the time and the CCT (the GAT is accurate at 520 μm).

Record: "Tn by GAT: RE 18, LE 16 mm Hg at 10 am."

Gonioscopy (card G9)

1. After tonometry, before dilatation.
2. Each quadrant: deepest structure seen and Shaffer grade (4 = 35–40°, 3 = 25–35°, 2 = 20°, 1 = 10°, **slit** < 10°, 0 = closed).



Deepest structure seen in each quadrant (SS = scleral spur) — open angles in all quadrants

Disc examination

1. Dilate after tonometry and gonioscopy; direct ophthalmoscope first.
2. Slit lamp, +90 dioptre (D) lens: cup, rim against the **ISNT rule** (thickest Inferiorly, then Superiorly, Nasally, Temporally), notch, haemorrhage, laminar dots; red-free light for retinal nerve fibre layer (RNFL) defects.

Record: in slide order (Short-case version).

Confrontation fields (card G10)

1. Face to face, 60 cm apart; his right eye fixes your left, other eyes closed.
2. Bring a finger in from the periphery, midway, in each quadrant; compare with your own field.

Record: "Fields (confrontation test): full in all quadrants."

Summary and complete diagnosis

Mr ___, a ___-year-old man on ___ drops for glaucoma for ___ years, presented for follow-up with gradual, painless, progressive defective vision, without haloes, trauma or steroids. On examination: visual acuity ___ / ___; tension by GAT ___ / ___ mm Hg at ___; VH grade ___; RAPD ___; lens ___; open angles, deepest structure ___; vertical CDR ___ / ___ with ___; confrontation fields ___.

eye → disease → type / stage / grade → aetiology → complications or status → other eye

- “Both eyes primary open-angle glaucoma with pseudophakia, with intraocular pressure under control on medical management.”
- “Right eye pseudoexfoliation glaucoma, severe stage, intraocular pressure not at target on maximal medical therapy; left eye pseudoexfoliation syndrome.”

Differential diagnosis

Condition	For	Against
Ocular hypertension (OHT)	IOP > 21 mm Hg twice; open angle	Normal disc, rim and field
NTG	Glaucomatous disc and field	IOP always < 22 mm Hg on diurnal curve; more haemorrhages, notching, PPA
PXF glaucoma	Open angle; high IOP	Flakes on pupil margin and lens; bigger 24-hour swings; often one eye
Pigmentary glaucoma	Wide-open angle	Young myopic men; Krukenberg spindle; mid-peripheral transillumination defects
Steroid-induced glaucoma	Open angle; raised IOP	Steroid use; posterior subcapsular cataract
Angle-recession glaucoma	Open angle; raised IOP	Trauma; recession on gonioscopy; one eye

What you must know

Definition

Glaucoma: a chronic, progressive optic neuropathy with irreversible nerve damage and matching field defects, with or without raised IOP. **POAG:** open, normal-looking angle, reduced outflow, no other eye disease; onset after 40 (juvenile 3–40 years). Normal IOP **10–21 mm Hg**.

Risk factors: raised IOP; age (prevalence doubles each decade over 40); Black race; first-degree relatives; myopia; thin CCT; diastolic perfusion pressure < 55 mm Hg; diabetes, hypertension, thyroid disease; vasospasm.

Classification

Grading of glaucomatous optic nerve head damage, with the Z value for target pressure:

Stage	Disc	Field (mean deviation, MD, decibels, dB)	Z
Suspect	—	—	0
Mild (early)	Minimal cupping	Nasal step or paracentral scotoma; MD better than -6 dB	1
Moderate	NRR thinning	Arcuate scotoma; MD -6 to -12 dB	3
Severe	Marked cupping	Within the central 5°; MD worse than -12 dB	5
End-stage	Gross cupping	Small residual field	7

Pathogenesis

- **Outflow resistance:** greatest at the juxtacanalicular trabecular meshwork.
- **Mechanical:** raised IOP bows the lamina cribrosa back → axoplasmic transport blocked → ganglion cell apoptosis.
- **Vascular** (key in NTG): vasospasm, nocturnal hypotension, blood loss.
- **Poles first** (larger laminar pores): cupping often starts inferotemporally → early **upper arcuate scotoma**.
- **Genes:** myocilin (chromosome 1), optineurin (chromosome 10).

Investigations

Test	Why	Expected finding
Diurnal phasing: IOP 3-hourly for 24 hours	Baseline, peak, drop timing; NTG versus POAG	Swing ≥ 8 mm Hg is significant (normal 3–6)
Pachymetry (CCT)	Correct the IOP reading	Thin < 500 μm reads falsely low; thick > 570 μm high
Humphrey 24-2 or 30-2	Stage and follow (card G10)	Paracentral, arcuate, nasal step; respects the horizontal meridian
OCT of RNFL; macular ganglion cell complex (GCC)	Pre-perimetric diagnosis; progression	Superior and inferior thinning; double hump lost; eyes differ by > 9 μm ; GCC thinning

Management

Target pressure: the IOP at or below which further damage is unlikely.

- **TP = IP (1 - IP/100) - Z $\pm$ 2** (TP target pressure; IP initial pressure; Z from the staging table).
- Example: IP 30 mm Hg, moderate (Z 3): $30 \times 0.7 - 3 = 18$, so **16–20 mm Hg**.

- Or lower IOP by **20%** (OHT), **30%** (mild damage, NTG), **40%** (advanced).

Medical: start one drug, a **PG analogue**; review at 2 months; if not at target, switch before adding.

Class (IOP fall)	Drug and strength	Frequency	Main side effect / contraindication
PG analogue (25–32%)	Latanoprost 0.005% Travoprost 0.004% Bimatoprost 0.01%, 0.03%	Once, at bedtime	Hyperaemia, long lashes, iris pigmentation; avoid in uveitis
Beta-blocker (20–30%)	Timolol 0.25%, 0.5% Betaxolol 0.5% (cardioselective)	Twice daily	Bronchospasm, bradycardia; not in asthma, COPD, heart failure, heart block
Alpha-2 agonist (15–20%)	Brimonidine 0.1%, 0.15%, 0.2%	Three times, sometimes twice	Allergy, dry mouth, somnolence; not under 2 years
CAI (15–20%)	Dorzolamide 2% Acetazolamide 250 mg tablet	Three times 2–4 times	Stinging; tablets: paraesthesia, renal stones; not in sulpha allergy
Miotic (15–20%)	Pilocarpine 0.5–4%	2–4 times	Browache, induced myopia; not in uveitic or neovascular glaucoma

Laser trabeculoplasty, selective (SLT) or argon (ALT): when drops cannot be used reliably (cost, memory, instillation, intolerance). Suits POAG, pigmentary, exfoliation, steroid-induced glaucoma; not uveitic, neovascular or angle-recession glaucoma.

Trabeculectomy indications (card G2): not at target on **3 drugs**, or fields progressing on maximal therapy; side effects or poor compliance; baseline IOP > **40 mm Hg**; one eye lost; difficult follow-up.

Cataract with glaucoma: expect a miotic pupil, synechiae, weak zonules (PXF), bleeding, a postoperative IOP rise.

- **Cataract alone:** IOP controlled on 1–2 drugs, no significant damage; risks an early IOP spike; later filtering surgery fails more.
- **Combined with filtering surgery:** many drugs, poor compliance, significant field loss, PXF, one-eyed. More complications, slower recovery.

Follow-up: IOP and compliance 3-monthly (mild–moderate) or monthly (advanced); fields 6-monthly or 3-monthly; gonioscopy yearly (a thickening lens can add angle closure).

Complications

- **Disease:** progressive field loss; blindness; sudden loss (acute IOP rise, vein occlusion, snuff-out).
- **Treatment:** surface toxicity, drug side effects; long-term drops lower surgical success.

Prognosis and counselling

- Damage is **irreversible**; treatment stops further loss. Central vision goes last.
- Chronic: drops and reviews for life; check compliance every visit.
- **Drop technique:** PG at bedtime, timolol in the morning; after each drop, lacrimal occlusion (press the inner corner) with eyes closed for **3 minutes**.
- Examine first-degree relatives (high risk).

Recent advances

- **SLT:** frequency-doubled neodymium-doped yttrium aluminium garnet (Nd:YAG) 532 nm laser targeting pigmented trabecular cells.
- **Microinvasive glaucoma surgery (MIGS):** Schlemm's canal stents, goniotomy, canal dilatation, suprachoroidal stents.

Viva questions

Why did you ask / examine...?

Q Why did you ask about night blindness?

Dark adaptation is defective in open-angle glaucoma; miotics worsen it.

Q Why did you ask about frequent change of glasses?

Pressure on the zonules and ciliary body impairs accommodation; field loss is mistaken for poor vision.

Q Why did you ask about haloes and headache with vomiting?

To rule out angle closure; haloes mean corneal oedema.

Q Why did you ask about sudden loss of vision?

Acute IOP rise, central retinal vein occlusion, or snuff-out after trabeculectomy.

Q Why did you ask about asthma, heart disease and sulpha allergy?

No beta-blocker in asthma, COPD, heart failure or heart block; no CAI in sulpha allergy.

Q Why did you measure tension before gonioscopy and dilatation?

Gonioscopy lowers it; dilatation raises it by 4–5 mm Hg.

Diagnosis

Q What makes you suspect glaucoma?

Raised IOP; vertical CDR ≥ 0.7 or asymmetry > 0.2; rim breaking ISNT; RNFL defect; field defect.

Q Why is the vertical cup-disc ratio more important?

Early rim loss is at the poles; a normal cup is horizontally oval.

Q What is the significance of a disc haemorrhage?

Active disease: flame-shaped, inferotemporal, precedes field loss by months; commoner in NTG.

Q What are bayoneting, nasalisation and baring of circumlinear vessels?

Double angulation of a vessel; vessels pushed nasally; a vessel left bare as the rim recedes.

Q What are the laminar dot sign and beta-zone atrophy?

Laminar pores seen in an advanced cup; atrophy next to the disc with sclera visible.

Q How does NTG differ from POAG?

A diagnosis of exclusion: IOP below 22 mm Hg on diurnal testing; more haemorrhages, notching and PPA.

Q When will you suspect a lesion other than glaucoma?

Pallor more than cupping, dyschromatopsia, a field respecting the vertical meridian: order neuroimaging.

Investigations

Q What is the role of CCT?

The GAT is accurate at 520 µm; thin corneas read falsely low, thick ones high.

Q Why do diurnal phasing?

Baseline and peak; progression despite good readings; drop timing; NTG versus POAG.

Q Can the field be normal with a glaucomatous disc?

Yes: 25–30% of ganglion cells are lost before white-on-white perimetry shows it.

Q How does glaucoma show on OCT?

Superior and inferior RNFL thinning (7 o'clock right eye, 5 o'clock left first); flattened double hump.

Management and surgery

Q How do you calculate target pressure?

$TP = IP (1 - IP/100) - Z \pm 2$; Z 0 suspect, 1 early, 3 moderate, 5 severe, 7 end-stage.

Q Why is a PG analogue given at bedtime?

Its effect peaks 10–14 hours later, with fewer vascular-dilatation symptoms.

Q Why is timolol best given in the morning?

It blunts the early-morning rise and avoids hypotension in sleep.

Q When will you advise trabeculectomy?

Not at target on 3 drugs or progressing; poor compliance or side effects; baseline > 40 mm Hg; one eye lost.

Complications

Q What are the systemic effects of timolol, and how do you reduce them?

Bradycardia, hypotension, bronchospasm, depression; lacrimal occlusion with eyes closed for 3 minutes.

Recent advances

Q What is SLT?

A 532 nm frequency-doubled Nd:YAG laser targeting melanin in trabecular cells.

Q What is MIGS?

Minimal-tissue pressure-lowering surgery for mild to moderate glaucoma: canal stents, goniotomy, suprachoroidal stents.

Examiner traps

- IOP without method, time and CCT, or after gonioscopy or dilatation.
- "Open angle" without gonioscopy; reversing Scheie and Shaffer (Scheie 4: only Schwalbe's line visible; Shaffer 4: wide open).
- "CDR 0.8" alone: give the vertical ratio, rim (ISNT), notch, haemorrhage, vessels, PPA, RNFL.
- Missing PXF at the pupil or lens: it changes diagnosis and cataract surgery.
- CCT in the clinical table: it is an investigation.
- Pallor more than cupping called glaucoma: needs neuroimaging.

Short-case version

Spot diagnosis: "Glaucomatous optic disc: vertically enlarged cup with neuroretinal rim thinning, suggestive of glaucomatous optic neuropathy."

Focused examination:

1. Pupils for RAPD before dilating.
2. Direct ophthalmoscope at arm's length (red glow), then close: media, disc.
3. Slit lamp with +90 D: cup, rim (ISNT), notch, haemorrhage, laminar dots; red-free light for RNFL defects.
4. Describe in slide order: size, disc shape, cup shape, margins, colour, cup-disc ratio, rim, vessels, arteriovenous ratio, peripapillary changes, RNFL defects.

Model: "Disc normal in size, vertically oval; cup vertically oval; margins well defined; pale; cup-disc ratio 0.9; circumferential rim thinning; centrally arising, dichotomously branching vessels; arteriovenous ratio 2:3; peripapillary atrophy; superior and inferior wedge-shaped RNFL defects; suggestive of glaucomatous optic neuropathy."

Three signs to show:

1. Vertical cup-disc ratio ≥ 0.7 , or asymmetry > 0.2 .
2. Rim thinning or a **notch** breaking the ISNT rule.
3. **Bayoneting** and **nasalisation**.

Say-it script

SAY IT

My patient is a 70-year-old man on regular follow-up, on eye drops for glaucoma in both eyes for 2 years. He has gradual, painless, progressive defective vision in both eyes, with no coloured haloes, headache with vomiting, redness, trauma or steroid use.

Best-corrected visual acuity is 6/18 in the right eye and 6/9 in the left. Intraocular pressure by Goldmann applanation tonometry is 18 and 16 millimetres of mercury at ten in the morning. The anterior chambers are Van Herick grade four, there is no relative afferent pupillary defect, and both eyes are pseudophakic. Gonioscopy shows open angles, with the scleral spur seen in all four quadrants. The right disc has a vertical cup-disc ratio of 0.8 with superior and inferior rim thinning, nasalisation, bayoneting and peripapillary atrophy; the left has 0.6 with an inferior notch. Confrontation fields are full.

My diagnosis is both eyes primary open-angle glaucoma with pseudophakia, with intraocular pressure under control on medical management.

I would do pachymetry, Humphrey fields, optical coherence tomography and diurnal phasing, and set a target pressure. If it is met, I continue the prostaglandin analogue; if not, I switch or add a drug, then consider laser trabeculoplasty or trabeculectomy. I will counsel him on lifelong treatment, compliance and drop technique.

Quick recall

- Normal IOP **10–21 mm Hg**; asymmetry ≥ 4 mm Hg is significant.
- Vertical CDR ≥ 0.7 or asymmetry > 0.2 : suspect glaucoma.
- **ISNT**: rim thickest Inferiorly, then Superiorly, Nasally, Temporally.
- **25–30%** of ganglion cells are lost before a field defect.
- **TP = IP (1 – IP/100) – Z $\pm$ 2**; Z = 0, 1, 3, 5, 7.
- PG analogue first, at bedtime; timolol in the morning.

Mnemonic: PART B before dilating: Pupillary reflex, Angle, Rubeosis, Tension, BCVA (best-corrected visual acuity).

Read more: Baidya p.152–167, 195–198 · Namrata p.171–178 · Aravind 4.2, 4.4, 4.8, 4.15, glaucoma model case sheet p.661–667 · Glaucoma case format p.1–2

G4 Primary angle-closure disease — PACS, PAC and PACG, including after laser iridotomy

★ Kept 4 times at JEH last year (one as the long case) · Long case

At a glance

- **Primary angle closure:** the peripheral iris closes the angle, usually by **pupillary block**; staged as primary angle-closure suspect (PACS), primary angle closure (PAC) or primary angle-closure glaucoma (PACG); often seen after laser peripheral iridotomy (LPI).
- Kept: bilateral, stable, full of signs — shallow anterior chamber (AC), iridotomies, glaukomflecken, iris atrophy. Gonioscopy was the listed task in all 4.
- Examiners test: Van Herick grade, gonioscopy **with indentation**, the stage, the mechanism, imaging, LPI, the acute attack.

Case format — history

Chief complaint: “Mrs X, a 58-year-old woman, has come for regular follow-up for glaucoma, after laser to both eyes 3 years ago.”

House wording: “Patient was apparently normal ___ years ago, after which she had episodes of coloured haloes and brow ache in dim light; laser was done to both eyes and she was started on timolol 0.5% eye drops twice daily.”

What to ask	Why it matters
History of present illness: haloes, brow ache, “smoke-filled room” blurring — in the dark, reading, after dilating drops, with stress	Intermittent closure: a mid-dilated pupil gives maximum pupillary block
Sudden painful loss of vision with redness, headache, vomiting — or gradual painless blurring	Acute attack (now or past) versus chronic closure, mostly without symptoms
Laser and surgery (which eye, when); drops, compliance; old intraocular pressure (IOP) and field records	Status after LPI; control; progression
Use of glasses	Hypermetropia: short eye, thick lens
Negative history: no coloured haloes	Rules out intermittent closure ; haloes also occur in incipient cataract (Fincham test)
No painful red, watering eye with headache and vomiting	Rules out an acute congestive attack and its mimics: glaucomatocyclitic crisis , uveitic glaucoma, herpetic keratouveitis
No trauma, no recent eye surgery	Rules out traumatic angle closure and malignant glaucoma
No diabetic eye disease, vein occlusion or retinal laser	Rules out neovascular glaucoma and ciliary body engorgement
No topiramate or sulpha drugs	Rules out drug-induced ciliary effusion
Past, drug and surgical: diabetes, hypertension	Repeated dilatation can start an attack
Asthma, heart or kidney disease; drug allergy (sulpha)	Choice of drugs (asthma: no beta-blocker)
Anticholinergic or adrenergic drugs (antihistamines, antidepressants, phenylephrine)	Precipitate angle closure
Personal and family: age, sex; glaucoma in the family	Middle-aged and older women ; first-degree relatives at high risk

Case format — examination

General and systemic (this case only)

- Before drugs: wheeze (no beta-blocker), blood sugar (glycerol), weight (doses per kilogram).

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Hypermetropic correction	6/12; +2.00 D sph (diopetre sphere): 6/9
Conjunctiva	Attack: circumciliary congestion ; on drops: inferior congestion	Quiet
Cornea	Attack: epithelial and stromal oedema ; chronic: Krukenberg spindle	Clear
Anterior chamber	Shallow ; Van Herick (VH) grade ; cells and flare = recent attack	Shallow, VH grade II
Iris	Iridotomy (slide order); sectoral atrophy , whorling; posterior synechiae; iris bombé in an attack	Patent LPI at 11 o'clock
Pupil	Attack or old attack: mid-dilated, vertically oval, poorly reacting	RTL 3 mm (round, reacting to light)

Structure	Look for in this case	How to record (example)
Lens	Glaukomflecken (grey-white anterior subcapsular flecks); thick or intumescent lens	Glaukomflecken +
Intraocular pressure	Goldmann applanation tonometer (GAT) before gonioscopy or dilatation , with time; attack 50–100 mm Hg	Tn (tension) by GAT 16 mm Hg at 10 am
Gonioscopy	Static, then indentation: iridotrabecular contact (ITC), peripheral anterior synechiae (PAS)	Cross diagram (below)
Fundus	No dilatation before iridotomy : direct or +90 D; attack: hyperaemic, oedematous disc; chronic: cupping	Undilated, +90 D: CDR (cup–disc ratio) 0.7
Fields	Confrontation	Full in all quadrants

Special tests

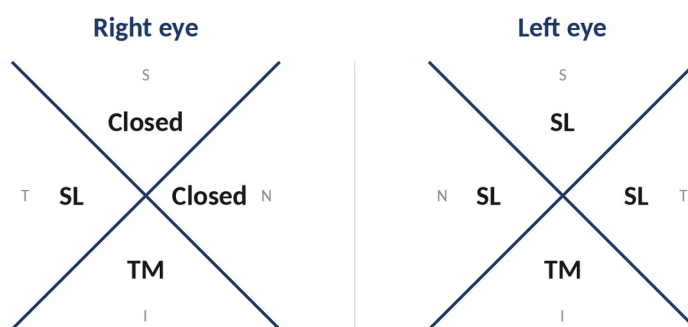
Anterior chamber depth (department script)

1. Pen-torch held **temporally**: whole iris lit = **grade 4**, normal depth; 2/3 lit = **grade 2**, shallow.
2. Van Herick: **slit-thin beam**, **1 mm inside the temporal limbus**, at **60°**; ratio of corneal thickness to the cornea–iris gap.
3. Grades (peripheral depth ÷ corneal thickness): **4** ≥ 1 (1:1, normal depth); **3** ¾ to ½; **2** ¾ (1:1¼); **1** < ¾; **0** iridocorneal contact. Grades 0–2: suspicious of angle closure. Depth < ¾ corneal thickness: the angle may be occludable.

Record: “Slit-thin slit 1 mm inside the temporal limbus at 60°; ratio 1:1¼; Van Herick grade 2, suggestive of shallow anterior chamber with occludable angles.”

Gonioscopy (technique: card G9)

1. Both eyes, before dilatation. Corneal oedema: wait, or clear it with **anhydrous glycerin**.
2. **Static**: Goldmann **two-mirror** lens without pressure (pressure distorts the angle). Posterior trabecular meshwork (TM) not seen = **ITC**; count the quadrants.
3. **Indentation**: Zeiss **four-mirror** lens, press: angle opens = **appositional**; stays shut = **synechial** (PAS; note extent). **Double hump** = plateau iris.



Deepest structure seen, before indentation — iridotrabecular contact in 3 quadrants in each eye (SL = Schwalbe's line)

Record: “Right eye (RE): ITC in 3 quadrants; indentation opens temporal to scleral spur (appositional); superior and nasal PAS. Left eye (LE): ITC in 3 quadrants, all opening on indentation.”

Iridotomy (slide order)

1. **Margin**: ragged, irregular = laser (surgical iridectomy: sharp, well-defined).
2. **Site**: clock hour.
3. **Surrounding iris**: hypopigmented, with pigment dispersion.
4. **Retro-illumination test for patency**: light reflex seen = patent.

Record: “Iridotomy at 11 o'clock, ragged irregular edges, hypopigmented surrounding iris, light reflex on retro-illumination — suggestive of a patent laser peripheral iridotomy.”

Summary and complete diagnosis

“Mrs ___, a ___-year-old hypermetropic woman, presented with ___. Visual acuity ___ / ___; AC shallow, VH grade ___ / ___; iridotomies ___; Tn by GAT ___ / ___ mm Hg at ___. Gonioscopy: ITC in ___ quadrants, PAS ___. CDR ___ / ___. Fields ___.”

eye → stage (PACS / PAC / PACG) → mechanism → acute or chronic; LPI patent or not → IOP control and treatment → other eye

- “Both eyes primary angle-closure glaucoma, status post laser peripheral iridotomy with patent iridotomies, with intraocular pressure under control on one medication (timolol 0.5% twice daily).”
- “Both eyes primary angle-closure suspect: iridotrabecular contact in 3 quadrants, no peripheral anterior synechiae, normal intraocular pressure, discs and fields; not on treatment.”

Differential diagnosis

Condition	For	Against
Primary open-angle glaucoma	Raised IOP, cupping	Open angle ; normal AC depth
Plateau iris syndrome	Narrow angle, attacks	Deep central AC ; flat iris; double hump ; still occludable after a patent LPI; younger
Lens-induced closure	Shallow AC, high IOP	Intumescent or subluxated lens
Malignant glaucoma	Shallow AC, high IOP	After surgery; diffuse shallowing; ultrasound biomicroscopy (UBM): ciliary processes rotated forward
Neovascular glaucoma	Closed angle, high IOP	New vessels on iris or angle

Primary angle closure tends to be bilateral: a **wide-open angle in the fellow eye** points to another diagnosis.

What you must know

Definition

Primary angle closure: appositional or synechial closure of the angle caused by **pupillary block**, with or without raised IOP or optic neuropathy, acute or chronic; secondary closure is excluded.

Classification

The **ISGEO classification**, based on IOP, gonioscopy, disc and fields:

Stage	Gonioscopy	IOP and PAS	Disc and field
PACS	ITC in ≥ 3 quadrants (almost 270°)	Normal IOP, no PAS	Normal
PAC	ITC in ≥ 3 quadrants	Raised IOP and/or PAS	Normal
PACG	ITC in ≥ 3 quadrants	Raised IOP and/or PAS	Glaucomatous optic neuropathy : cupping, field loss

Mechanism — four sites: iris (**pupillary block**, most common) · ciliary body (**plateau iris**) · lens (thick, intumescent or subluxated) · behind the iris (“pushing”: **malignant glaucoma**, cysts). Secondary “pulling”: neovascular glaucoma, iridocorneal endothelial syndrome.

Gonioscopic grading — Shaffer:

Grade	Width	Deepest structure seen	Closure
4 wide open	35–40°	Ciliary body	Impossible
3 open	25–35°	Ciliary body	Impossible
2 moderately narrow	20°	Pigmented TM	Possible but unlikely
1 extremely narrow	10°	Only Schwalbe’s line; maybe anterior TM	Possible
Slit	< 10°	None; no obvious iridocorneal contact	Portions appear closed
0 closed	—	Iridocorneal contact	Complete

Occludable angle: pigmented TM not visible (Shaffer 1 or 0) without indentation in at least **three quadrants**.

Spaeth — four parameters: iris-root insertion **A–E** · angle width **10–40°** · peripheral iris contour **S, R, Q** · TM pigmentation **0–4**.

Pathogenesis

- **Relative pupillary block:** lens–iris touch traps aqueous in the posterior chamber, bowing the peripheral iris onto the TM; worst at **mid-dilatation** (dim light).
- **Anatomy:** short axial length, small cornea, thick lens and anteriorly placed iris–lens diaphragm; the lens thickens and moves forward with age.
- Contact is first **appositional**, then **synechial**; outflow falls and IOP rises. Above **40 mm Hg** the sphincter is ischaemic; infarcted lens fibres form glaukomflecken.

Investigations

Test	Why	Expected finding
Gonioscopy with indentation	Gold standard ; stage; appositional versus synechial	ITC ≥ 3 quadrants; PAS
Anterior segment optical coherence tomography (AS-OCT)	Non-contact, quick, 360°; lens vault	Occludable angle; cannot see behind the pigmented iris epithelium

Test	Why	Expected finding
UBM (50–100 MHz)	Mechanism: ciliary body, sulcus, zonules; works through corneal oedema	Iris bowing (block); double hump on indentation (plateau iris); cysts
Perimetry; central corneal thickness (after an attack)	Stage (PAC versus PACG); applanation accuracy	Glaucomatous field loss

Management

By stage: PACS — **LPI**, or observation with serial gonioscopy; PAC and PACG — **LPI plus medical or surgical** IOP control.

LPI in PACS if: frequent dilatation (diabetes, hypertension, age-related macular degeneration), an attack in the other eye, hypermetropia, intermittent symptoms, predisposing drugs, unreliable follow-up.

Neodymium:yttrium-aluminium-garnet (Nd:YAG) iridotomy (photodisruption; all iris colours)

1. **Pilocarpine 1%**; brimonidine or apraclonidine half an hour before; **proparacaine 0.5%**.
2. **Abraham lens** (+55 D button): steadies the eye, magnifies, concentrates energy.
3. **Site:** peripheral third, in a **crypt**, **11–1 o'clock** under the upper lid.
4. **2–5 mJ, 1–3 pulses**; optimal size **150–200 µm**.
5. **End point:** sudden gush of aqueous or pigment; the chamber **deepens**.
6. **After:** IOP at **1 hour** (spike: rise ≥ 8 mm Hg or IOP > 30 mm Hg); **prednisolone acetate 1% four times a day for 5–7 days**; **gonioscopy at 1 week**.

Argon (thermal): anticoagulated patients; pre-treatment of thick dark irides. No LPI with corneal oedema, a flat AC, a totally synechial angle, uveitis or neovascular glaucoma.

Acute attack — step by step

1. **Mannitol 20% intravenously, 1–2 g/kg over 20–30 minutes:** miotics fail while IOP is very high.
2. If no nausea or vomiting: **acetazolamide 250–500 mg orally stat, then 250 mg four times a day**, or **glycerol 50% orally, 1–1.5 g/kg** (not in diabetes); **apraclonidine 0.5%**.
3. IOP **below 40 mm Hg:** **pilocarpine 2%** four times in 30 minutes, then 6-hourly.
4. **Timolol 0.5% twice daily** with **brimonidine 0.2% three times daily**.
5. **Prednisolone acetate 1% four times a day; ibuprofen 500 mg twice daily;** antiemetic.
6. Review **hourly**; gonioscopy of both eyes once the cornea clears. Indenting the central cornea may open the angle.
7. Quiet eye, clear cornea: **Nd:YAG iridotomy in both eyes**.
8. Hazy cornea, closure $< 2/3$: **surgical iridectomy** (YAG in the fellow eye). Closure $> 2/3$ or persistent inflammation: **primary trabeculectomy**.

Plateau iris syndrome: LPI first; if still occludable, **argon laser iridoplasty**.

Medical after LPI: as in primary open-angle glaucoma (card G1).

Trabeculectomy (5-fluorouracil or mitomycin C): IOP uncontrolled despite LPI and drugs; progression on maximal tolerated therapy; **synechial closure $> 270^\circ$** . Non-penetrating surgery is relatively contraindicated.

Lens extraction

- **Phacoemulsification with intraocular lens implantation** corrects hypermetropia, deepens the anterior chamber and opens the angle: IOP control in almost all patients with normal preoperative IOP, and in up to **80%** with raised preoperative IOP. (*Kanski — not in your practical books*)
- **EAGLE study:** clear lens extraction is more effective and more cost-effective than LPI in PAC with IOP > 29 mm Hg, or in PACG. (*Kanski — not in your practical books*)
- **PACS — ZAP trial:** LPI has a modest prophylactic effect; offer it only to those at highest risk of PACG. If significant ITC persists after LPI and there is symptomatic cataract, lens extraction usually opens the angle. (*Kanski — not in your practical books*)

Fellow eye: prophylactic LPI early. **Follow-up:** serial gonioscopy; high IOP with an open angle after LPI = **combined mechanism:** treat the open-angle part.

Complications

- **Of an attack:** glaukomflecken, iris atrophy, PAS, posterior synechiae, endothelial loss, cupping; **absolute glaucoma**.
- **Of LPI:** IOP spike, uveitis, hyphaema, focal cataract, glare; rarely **malignant glaucoma** (treated by Nd:YAG hyaloidotomy).

Prognosis and counselling

- PACS $\rightarrow$ acute closure **6–10%**; PAC or PACG **17–35%**. Fellow eye: angle closure in about **50% within 5 years**.
- A patent LPI is not a cure (plateau iris, PAS, combined mechanism): lifelong review.
- Counsel: no dilating or anticholinergic drugs before iridotomy; report haloes or pain at once; examine relatives.

Recent advances

- AS-OCT may detect occludability better than gonioscopy, but gonioscopy remains the **gold standard**; imaging is complementary, and UBM is still needed behind the iris.

Viva questions

Why did you ask / examine...?

Q Why did you ask about coloured haloes?

Corneal epithelial oedema acts as a diffraction grating (blue-green centre, yellow-red outside). Also: incipient cataract, mucus, vitreous opacities, tilted intraocular lens.

Q Why did you ask about diabetes and hypertension?

Repeated dilatation can start an attack: an indication for LPI.

Q Why did you ask about drugs?

Anticholinergics and adrenergics precipitate angle closure; topiramate and sulpha drugs cause ciliary effusion.

Q Why did you not dilate the pupil?

Contraindicated until iridotomies are done; I used a +90 D lens.

Q Why did you look for glaukomflecken?

A sign of a previous attack. **Vogt's triad:** glaukomflecken, sectoral iris atrophy, Krukenberg spindle.

Diagnosis

Q Define PACS, PAC and PACG.

All: ITC in ≥ 3 quadrants. PACS: normal IOP, disc and fields, no PAS. PAC: raised IOP and/or PAS, normal disc. PACG: plus glaucomatous optic neuropathy.

Q Plateau iris configuration versus syndrome?

Configuration: anterior ciliary processes hold the peripheral iris forward, without closure. Syndrome: the angle stays occludable despite a patent LPI.

Q Why is the pupil mid-dilated and fixed in an attack?

Above 40 mm Hg the sphincter is ischaemic; pain adds sympathetic overactivity.

Q Why is pain worse than in open-angle glaucoma?

Pain follows the speed of the rise, not the level.

Q What is the Fincham test?

A stenopaic slit moved across the pupil: the glaucoma halo stays intact; the cataract halo breaks into colours.

Investigations

Q AS-OCT or UBM?

AS-OCT: non-contact, quick, 360°; cannot see behind the pigmented iris epithelium. UBM: 50–100 MHz, axial resolution 25 μm , 4–5 mm deep; sees the ciliary body, even through corneal oedema.

Q How are UBM and AS-OCT findings described?

UBM: echogenic, hypoechoic, echolucent; radial scan for the angle, transverse for clock-hour extent. AS-OCT: hyperreflective, hyporefective.

Q Name the provocative tests.

Mydriatic, dark room (90 minutes), prone (45–60 minutes), prone dark room, phenylephrine–pilocarpine, triple. Mydriatic or dark room positive: rise > 8 mm Hg or closure on gonioscopy.

Management and surgery

Q Why does iridotomy work?

Aqueous bypasses the pupil; the gradient goes, the iris flattens and the chamber deepens.

Q Why Nd:YAG rather than argon?

Photodisruption works on all iris colours, with fewer late closures; argon needs pigment.

Q Why place the iridotomy at 11–1 o'clock?

The upper lid covers it; the upper nasal iris avoids diplopia and macular burns.

Q Why does lens extraction help in angle closure?

The lens thickens and moves forward with age. Phacoemulsification with intraocular lens implantation corrects hypermetropia, deepens the anterior chamber and opens the angle; in the EAGLE study, clear lens extraction was more effective than LPI in PAC with IOP > 29 mm Hg, or PACG. (*Kanski — not in your practical books*)

Q When would you do trabeculectomy?

IOP uncontrolled despite LPI and drugs, progression, synechial closure $> 270^\circ$, or an attack with closure $> 2/3$.

Q Why repeat gonioscopy after LPI?

To prove the block is relieved (lens capsule seen through the iridotomy), map PAS and unmask plateau iris.

Complications

Q Complications of LPI?

IOP spike, uveitis, hyphaema, focal cataract, posterior synechiae, glare; rarely malignant glaucoma.

Recent advances

Q Newer imaging for angle closure?

AS-OCT and UBM. AS-OCT may detect occludability better than gonioscopy, but gonioscopy remains the gold standard.

Examiner traps

- Dilating before gonioscopy and iridotomy.
- “Patent” on retro-illumination alone: the lens capsule must be seen through it.
- Pressing during static grading, or skipping indentation.
- Staging without disc and fields: PAC and PACG differ only by optic neuropathy.
- Pilocarpine first when IOP is above 40 mm Hg.
- Forgetting the fellow eye.

Say-it script

SAY IT

My patient, a fifty-eight-year-old hypermetropic woman, has come for regular follow-up. Three years ago she had evening haloes and brow ache with raised pressure; laser was done to both eyes, and she uses timolol twice daily. She has no history of a painful red eye with vomiting, trauma, eye surgery, sulpha drugs or asthma.

Best-corrected visual acuity is 6/9 in each eye. The anterior chambers are shallow, Van Herick grade two. Each eye has an iridotomy at eleven o'clock with ragged edges, hypopigmented surrounding iris and a clear light reflex on retro-illumination, suggestive of patent laser peripheral iridotomies. Intraocular pressure by Goldmann applanation is 16 and 18 millimetres of mercury at ten in the morning. Gonioscopy shows iridotrabecular contact in three quadrants in each eye; on indentation the angles open, except superiorly and nasally in the right eye, where there are peripheral anterior synechiae. The undilated discs show cup-disc ratios of 0.7 and 0.6 with inferior rim thinning. Confrontation fields are full.

My diagnosis is both eyes primary angle-closure glaucoma, status post laser peripheral iridotomy with patent iridotomies, with intraocular pressure under control on one medication.

I would confirm with automated perimetry, look for plateau iris with ultrasound biomicroscopy, continue timolol, repeat gonioscopy at each review, and consider trabeculectomy if the disease progresses.

Quick recall

- ISGEO: **ITC in ≥ 3 quadrants**; PAC adds raised IOP and/or PAS; PACG adds optic neuropathy.
- Attack IOP **50–100 mm Hg**; pilocarpine only below **40 mm Hg**.
- Mannitol 20% **1–2 g/kg**; acetazolamide **250 mg** four times a day; glycerol **1–1.5 g/kg**.
- Nd:YAG LPI: **2–5 mJ**, **1–3 pulses**, **150–200 μm** ; spike = rise ≥ 8 or IOP **> 30 mm Hg**.
- Fellow eye **50% in 5 years**; PACS $\rightarrow$ PAC or PACG **17–35%**.
- Vogt's triad: glaukomflecken, sectoral iris atrophy, Krukenberg spindle.

Mnemonic: I CAN SEE TILL SCHWALBE'S LINE — Iris root, Ciliary body, Scleral spur, Trabecular meshwork, Schwalbe's line: the angle structures from the iris root forwards.

Read more: Baidya p.148–152, 441, 715 · Namrata p.178–184 · Aravind 4.2, 4.5, 4.6, 4.7, 4.17 · Kanski p.377

G5 Neovascular glaucoma, including rubeosis iridis

★ Kept once at JEH last year · Long case (rubeosis iridis is also a short case)

At a glance

- **Neovascular glaucoma (NVG):** a severe secondary glaucoma. A fibrovascular membrane covers the iris and angle after long-standing retinal ischaemia — **proliferative diabetic retinopathy (PDR)**, **ischaemic central retinal vein occlusion (CRVO)**, **ocular ischaemic syndrome**.
- Kept for its signs: neovascularisation of the iris (NVI) and of the angle (NVA), ectropion uveae, very high intraocular pressure (IOP). Last year's case had applanation tonometry as its task.
- Examiners test: iris and angle **before dilatation**; the vessels in slide order; **stage**; **cause**; anti-vascular endothelial growth factor (anti-VEGF) → panretinal photocoagulation (PRP) → surgery; drugs to avoid.

Case format — history

Chief complaint: “Mr X, a 62-year-old man, came with complaints of pain, redness and defective vision in the right eye for the past 10 days, and sudden loss of vision in the same eye 3 months ago.”

What to ask	Why it matters
History of present illness: pain, redness, haloes, headache, vomiting; bleeding into the eye	Acute angle-closure stage; hyphaema from fragile vessels
Earlier loss of vision — sudden or gradual?	Sudden: ischaemic CRVO (“ 100-day glaucoma ”), central retinal artery occlusion (CRAO). Gradual: diabetic retinopathy, chronic uveitis, long-standing retinal detachment, radiation retinopathy
Retinal laser, intravitreal injections; known glaucoma, drops, old records	Prior treatment changes the plan; pre-existing primary open-angle glaucoma (POAG) raises IOP more
Glasses for hypermetropia; similar attacks	Primary angle-closure glaucoma (PACG); chronic angle closure can itself cause NVG (important in India)
Negative history: no trigger (dim light, reading, dilating drops, stress)	Against primary acute angle closure
No recurrent red eye treated with steroid drops	Rules out uveitic glaucoma
No injury	Rules out old trauma (angle recession)
Past, drug and surgical: diabetes	Commonest cause of NVI (a third of rubeotic eyes)
Hypertension, hypercoagulable states	Cause of the ischaemia
Carotid or heart disease	Ocular ischaemic syndrome ; heart disease bars intravenous mannitol
Asthma; sulpha allergy	No beta-blocker; no carbonic anhydrase inhibitor (CAI)
Cataract surgery (capsule rupture, aphakia), vitrectomy	Lens, capsule and vitreous are barriers to angiogenic factors

Case format — examination

General and systemic (this case only)

- **Blood pressure, pulse, heart:** causes; rule out low blood pressure and heart disease before intravenous mannitol.
- **Diabetes, hypertension:** complications. **Chest:** asthma rules out beta-blockers.

Ocular examination — RE | LE template

Two columns: right eye (RE), left eye (LE). Pupil, iris, angle and tension **before any drop**.

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Poor, even no perception of light; corneal oedema adds blur	HM (hand movements)
Lids and conjunctiva	Lid oedema; circumciliary congestion	Circumciliary congestion +
Cornea	Epithelial and stromal oedema	Epithelial oedema +
Anterior chamber	Van Herick (VH) grade; flare , cells, hyphaema	VH grade ____; flare +
Iris	NVI in slide order; flat, glistening surface; peripheral anterior synechiae (PAS)	NVI 11 to 8 o'clock
Pupil	Mildly dilated, oval, fixed ; ectropion uveae ; relative afferent pupillary defect (RAPD)	Oval, fixed; ectropion uveae +; RAPD +
Lens	Cataract; pseudophakia with capsule dehiscence, or aphakia	PCIOL (posterior chamber intraocular lens)

Structure	Look for in this case	How to record (example)
Intraocular pressure	Goldmann applanation tonometry (GAT) ; with corneal oedema, take IOP as in a higher range	Tn (tension) by GAT 48 mm Hg at 10 am
Gonioscopy	NVA crossing the scleral spur; PAS extent	Cross diagram
Fundus	Dilate with a cycloplegic (also treats); +90 dioptre (D) lens. Vitreous haemorrhage; disc hyperaemia, oedema (acute) or pallor, cupping (chronic); the cause : CRVO, PDR, CRAO	Disc pale, CDR (cup–disc ratio) 0.8; suggestive of old CRVO

Special tests

1. New vessels on the iris — before dilatation (department slide)

1. No drops; slit lamp at **high magnification**: pupillary margin first, then the iris surface and any iridotomy margin.
2. Describe: **appearance → character → course → arrangement → extent** (clock hours).

Record: “RE: **thin, friable, arborising, irregularly arranged** new vessels at the pupillary margin, extending over the iris from **11 o'clock to 8 o'clock** — suggestive of neovascularisation of the iris.”

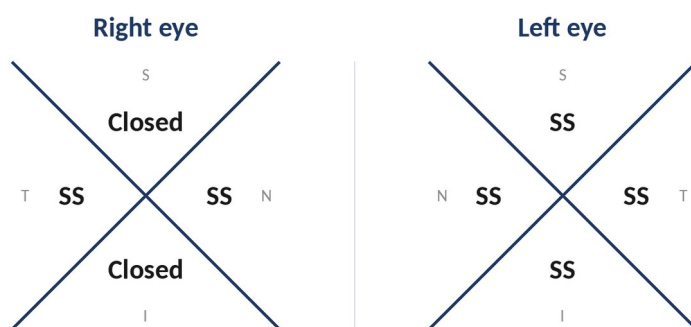
2. Applanation tonometry (card G8)

1. GAT in both eyes before dilatation; note the time.
2. Repeat once the corneal oedema clears.

Record: “Tn by GAT: RE 48, LE 14 mm Hg at 10 am.”

3. Gonioscopy (card G9)

1. Goldmann **2-mirror** lens, both eyes, before dilatation; defer while the cornea is oedematous.
2. NVA: vessels run from the iris root **across the ciliary body band and scleral spur**, arborising over the trabecular meshwork. Present without NVI in about **12%**.
3. PAS extent in degrees (PRP can still reverse the IOP rise if synechial closure is **not more than 270°**); **4-mirror** indentation separates appositional from synechial closure.



Deepest structure seen in each quadrant (SS = scleral spur) — RE: closed superiorly and inferiorly by PAS; new vessels cross the scleral spur temporally and nasally. LE: open angles in all quadrants

Summary and complete diagnosis

Mr ___, a ___-year-old man with diabetes / hypertension, presented with pain, redness and defective vision in the ___ eye for ___, after sudden / gradual loss of vision ___ months ago. That eye: vision ___; congestion, corneal oedema, flare; new vessels from ___ to ___ o'clock; ectropion uveae; RAPD ___; tension by GAT ___ mm Hg at ___; NVA ___, PAS over ___°; fundus ___. Other eye: ___.

eye → disease → stage → aetiology → complications or status → other eye

- “**Right eye neovascular glaucoma, angle-closure stage, secondary to ischaemic central retinal vein occlusion, with uncontrolled intraocular pressure; left eye normal.**”
- “**Left eye neovascular glaucoma, open-angle stage, secondary to proliferative diabetic retinopathy, with pseudophakia; right eye proliferative diabetic retinopathy, post panretinal photocoagulation.**”

Differential diagnosis

Condition	For	Against
Primary acute angle closure	Painful red eye, haloes, high IOP	No NVI, NVA, fibrovascular membrane or ectropion uveae
Uveitic glaucoma	Flare, cells, raised IOP; dilated iris vessels	Keratic precipitates; dilated vessels are not rubeosis
Fuchs' heterochromic iridocyclitis	Fine angle vessels, spontaneous hyphaema	White, quiet eye; stellate keratic precipitates; heterochromia
Iridocorneal endothelial (ICE) syndrome	PAS, distorted iris, raised IOP	Corneal decompensation, corectopia, pseudopolycoria

Condition	For	Against
Old trauma (angle recession)	PAS, distorted iris, hyphaema	Injury; recessed angle; no NVI
Normal iris vessels (light or atrophic iris)	Visible vessels	Stromal, radial, uniform, unbranched; no leak

What you must know

Definition

NVG: severe secondary glaucoma; fibrovascular tissue proliferates over the angle and iris, giving very high IOP, after long-standing, severe retinal ischaemia. **Rubeosis iridis:** new vessels on the iris surface, whatever the angle or IOP. Also called haemorrhagic, thrombotic, congestive or **100th-day glaucoma**.

Causes: the three commonest are **diabetic retinopathy**, **ischaemic CRVO** and **ocular ischaemic syndrome**; in India also **chronic angle-closure glaucoma**. Others: RUBEOTIC (Quick recall); also CRAO, sickle cell, Coats', cataract surgery, radiation.

Classification

Clinicopathological stages:

Stage	NVI and gonioscopy	IOP	Other signs
Prerubeosis	No NVI or NVA; open angle	Normal	Iris angiography shows early leakage
Pre-glaucoma (rubeosis iridis)	NVI at the pupil margin; open angle, early NVA	Normal	Usually none
Open-angle glaucoma	Florid NVI; open angle, florid NVA under a fibrovascular membrane	Raised	Cells, flare, hyphaema
Angle-closure glaucoma	Vessels less apparent; PAS up to total closure	Persistently high; can reach 60 mm Hg	Ectropion uveae, glistening iris, corneal oedema, disc damage

Some add a **burnt-out stage** (total synechial closure, few vessels).

Wand's classification: stage 1 vessels at the pupillary margin · 2 up to the collarette · 3 up to the angle · 4 crossing the scleral spur.

Pathogenesis

- **Retinal hypoxia** → angiogenic factors: **VEGF**, fibroblast growth factor, tumour necrosis factor- α , angiogenin.
- Lens, posterior capsule and vitreous are barriers (and supply vaso-inhibitory factors); capsule breach and vitrectomy remove them.
- Vessels bud from capillaries of the **minor arterial circle** at the pupillary margin; thin-walled and fenestrated, they leak and bleed.
- An invisible **fibrovascular membrane** covers the trabecular meshwork → **open-angle (pretrabecular) glaucoma**; its myofibroblasts contract → PAS, flat iris, **ectropion uveae** → **synechial angle closure**.

Investigations

Test	Why	Expected finding
Iris fluorescein angiography, fluorophotometry	Earliest sign, before vessels show	Leakage at the pupillary margin
Fundus fluorescein angiography (FFA)	Extent of ischaemia; plan PRP	Extensive capillary closure; leaking iris vessels
Fields; optical coherence tomography (OCT), nerve fibre layer	Known glaucoma	Progression
Systemic work-up	Cause; drug safety	Diabetes, hypertension, hypercoagulable state, heart or carotid disease

Management

Goals: control IOP, stop new vessels, treat the cause, relieve symptoms.

Step 1 — Medical. IOP below 40 mm Hg: topical aqueous suppressants. Above **40 mm Hg**: add intravenous mannitol.

Drug	Strength / dose	Frequency	Note
Timolol (beta-blocker)	0.5%	Twice daily	Not in asthma
Brimonidine (alpha-2 agonist)	0.2%	Three times	—
Dorzolamide (topical CAI)	2%	Three times	Not in sulpha allergy
Acetazolamide tablet	250 mg	Twice daily	Not in sulpha allergy
Mannitol 20% intravenous	1–2 g/kg over 20–30 min	Initial control	Not in heart disease, low blood pressure

Drug	Strength / dose	Frequency	Note
Prednisolone acetate	1%	6–8 times a day	Inflammation
Atropine (or homatropine 2%)	1%	Three times	Pain; cycloplegia is treatment

Avoid: miotics (more inflammation and pain), **prostaglandin analogues** and **adrenaline**.

Step 2 — Treat the cause once the cornea clears (gonioscopy and fundus are a must).

- **Bevacizumab 1.25 mg in 0.05 ml**, intravitreal through the pars plana (or intracameral, same dose). NVI regresses within **24–48 hours**, but the effect lasts only weeks, so it is an adjunct to PRP. Give it **2–3 days before** surgery: less bleeding, inflammation and failure.
- **PRP — the treatment of choice:** ablating ischaemic retina cuts VEGF; NVI regresses from 2 weeks, fully by **4–6 weeks**. It can still reverse the IOP rise while synechial closure is **not more than 270°**. Green 532 nm laser, **500 µm, 0.1 s**; mild white burns one burn-width apart, out to the equator; **1200–1600 burns in 3–4 sittings**.
- **Hazy media:** trans-scleral panretinal cryotherapy, or bevacizumab alone.

Step 3 — Surgery when IOP stays high after retinal ablation (extensive PAS).

1. **Visual potential** (rare): after anti-VEGF, **trabeculectomy with mitomycin C (MMC) 0.2–0.4 mg/ml for 2 min** (or 5-fluorouracil 50 mg/ml for 5 min). Poor alone; with pre-operative bevacizumab and/or PRP, success may reach **95%**.
2. **Glaucoma drainage device (Ahmed valve, Molteno, Baerveldt):** often **primary** when filtering surgery is likely to fail.
3. **Poor visual potential with pain: cyclodestruction** — diode or neodymium:yttrium-aluminium-garnet (Nd:YAG) laser cyclophotocoagulation, cyclocryotherapy.
4. **Blind, painful eye: retrobulbar alcohol injection**, or **enucleation** after counselling.

Follow-up: every visit — undilated pupil margin and angle at high magnification, IOP, disc; both eyes.

Complications

Hypaema, vitreous haemorrhage, optic atrophy. Late: **painful bullous keratopathy**, **complete synechial angle closure**, **intractable glaucoma**.

Prognosis and counselling

- Potentially blinding. Anti-VEGF with PRP controls IOP only in the open-angle stage; after extensive synechial closure, surgery is often needed and useful vision is rare.
- Control diabetes and hypertension.

Recent advances

- Intracameral bevacizumab alone can stabilise NVI and control IOP in the early open-angle stage.
- Newer: alpha-interferon, troxerutin, gene transfer of pigment epithelium-derived factor (PEDF).

Viva questions

Why did you ask / examine...?

Q Why did you ask about sudden loss of vision months earlier?

Ischaemic CRVO: NVG usually follows in 3–5 months (“100-day glaucoma”); range 2 months to 2 years.

Q Why did you ask about diabetes?

Diabetic retinopathy is the commonest cause of NVI: a third of rubeotic eyes have it.

Q Why did you ask about carotid and heart disease?

Ocular ischaemic syndrome is one of the three commonest causes; heart disease bars intravenous mannitol.

Q Why did you ask about cataract surgery or vitrectomy?

Lens, capsule and vitreous block angiogenic factors; removing them raises the risk.

Q Why did you look at the iris before dilating?

Early vessels are fine and at the pupillary margin; dilatation hides them.

Q Why gonioscopy when the iris is clean?

NVA can be present without NVI in about 12%.

Diagnosis

Q What is the first sign of NVG?

Clinically, vessel tufts at the pupillary margin; on fluorescein angiography, leakage there.

Q How do you tell new vessels from normal iris vessels?

New: superficial, irregular, varying size, branching, leaky, crossing the scleral spur. Normal: stromal, radial, uniform.

Q What is Wand’s classification?

Vessels at the pupillary margin (1), to the collarette (2), to the angle (3), crossing the scleral spur (4).

Q Why does ectropion uveae occur?

The contracting fibrovascular membrane pulls the posterior pigment layer round the pupil margin.

Q What closes the angle?

Myofibroblasts in the fibrovascular membrane contract: PAS and a flat iris.

Investigations

Q Why do new vessels leak fluorescein?

Gap junctions between endothelial cells and fenestrations in the basement membrane.

Q Where else does fluorescein leak from the pupillary margin?

Pseudoexfoliation (pupillary ruff defects) and Fuchs' heterochromic iridocyclitis.

Management and surgery

Q What is the treatment of choice, and how does it work?

PRP. Destroying retinal pigment epithelium and photoreceptors cuts oxygen demand, so fewer angiogenic factors are released.

Q What is the role of bevacizumab?

1.25 mg in 0.05 ml intravitreally; NVI regresses within 48 hours, so PRP and surgery bleed less. The effect lasts weeks.

Q Which drugs are contraindicated?

Miotics (more inflammation and pain), adrenaline, and prostaglandin analogues (more inflammation; the membrane blocks uveoscleral outflow).

Q Why atropine and a steroid?

They reduce inflammation and pain; cycloplegia is treatment even in angle closure.

Q Which operation would you choose?

Visual potential: trabeculectomy with MMC after anti-VEGF, or a drainage device. Poor potential: cyclophotocoagulation. Blind, painful: retrobulbar alcohol or enucleation.

Q When is PRP replaced by cryotherapy?

When hazy media prevent PRP: trans-scleral panretinal cryotherapy.

Q When will you do prophylactic PRP?

PDR; diabetic retinopathy with pupillary-margin leakage before lensectomy or vitrectomy; ischaemic CRVO with extensive ischaemia when follow-up is unreliable.

Complications

Q What are the late complications?

Painful bullous keratopathy, complete synechial angle closure, intractable glaucoma.

Recent advances

Q What newer treatments exist?

Anti-VEGF antibodies (bevacizumab), alpha-interferon, troxerutin, PEDF gene transfer.

Examiner traps

- Dilating, or any drop, before seeing the pupil margin and angle: early NVI is missed.
- No gonioscopy because the iris is clean: NVA occurs alone in 12%.
- Treating it as primary angle closure: miotics, prostaglandins and adrenaline are contraindicated.
- Lowering pressure without naming the cause: PRP is the treatment of choice; anti-VEGF lasts weeks.
- No stage or PAS extent: they decide whether PRP can still reverse the IOP rise (closure not more than 270°).
- Calling radial stromal vessels in a light iris "rubeosis".

Short-case version

Spot diagnosis: rubeosis iridis — new vessels at the pupillary margin and on the iris surface.

Focused examination:

1. Before any drop — **PART B:** Pupillary reflex, Angle, Rubeosis, Tension, Best-corrected visual acuity.
2. Slit lamp, **high magnification**, undilated pupil: pupillary margin, iris surface; describe in slide order.
3. **Gonioscopy:** NVA crossing the scleral spur; PAS extent.
4. **IOP** by GAT: normal IOP means the pre-glaucoma stage.
5. **Fundus** after dilatation for the cause: CRVO, PDR, CRAO.

Three signs to show:

1. New vessel **tufts at the pupillary margin**.
2. **Irregular, branching, non-radial** surface vessels.
3. **NVA crossing the scleral spur** (ectropion uveae when advanced).

"Right eye rubeosis iridis, pre-glaucoma stage, secondary to proliferative diabetic retinopathy."

Say-it script

SAY IT

My patient, a sixty-two-year-old man with diabetes and hypertension, came with pain, redness and defective vision in the right eye for ten days, with haloes and vomiting, three months after sudden loss of vision in that eye. There is no history of injury, eye surgery or steroid drops.

In the right eye, vision is hand movements, with circumciliary congestion, mild corneal oedema and flare. Before dilatation, I saw thin, friable, arborising, irregularly arranged new vessels at the pupillary margin from eleven to eight o'clock, with ectropion uveae and a relative afferent pupillary defect. Intraocular pressure by Goldmann applanation is forty-eight millimetres of mercury in the right eye and fourteen in the left at ten in the morning. Gonioscopy shows peripheral anterior synechiae superiorly and inferiorly, and new vessels crossing the scleral spur elsewhere. The fundus shows an old central retinal vein occlusion.

My diagnosis is right eye neovascular glaucoma, angle-closure stage, secondary to ischaemic central retinal vein occlusion, with uncontrolled intraocular pressure; left eye normal.

I would lower the pressure with timolol, brimonidine, dorzolamide, oral acetazolamide and intravenous mannitol, and give prednisolone acetate and atropine, avoiding miotics and prostaglandin analogues. Once the cornea clears, I would do fundus fluorescein angiography, give intravitreal bevacizumab, then panretinal photocoagulation. If the pressure stays high: a glaucoma drainage device or trabeculectomy with mitomycin C, or cyclophotocoagulation if the visual potential is poor.

Quick recall

- Commonest causes: **diabetic retinopathy, ischaemic CRVO, ocular ischaemic syndrome.**
- After CRVO: **"100-day glaucoma"** (usually 3–5 months).
- Vessels start at the **pupillary margin**; NVA without NVI in **12%**.
- **PRP**: treatment of choice; reverses the IOP rise only while synechial closure is **not more than 270°**; **1200–1600 burns**.
- **Bevacizumab 1.25 mg/0.05 ml**: NVI regresses in **24–48 hours**; **2–3 days** before surgery.
- IOP above **40 mm Hg**: mannitol 20%, 1–2 g/kg intravenously.
- **Avoid** miotics, prostaglandin analogues, adrenaline.

Mnemonic: RUBEOTIC (causes) — Retinopathy and Retinal vein occlusion (PDR, CRVO), Retinal detachment • Uveitis • Branch retinal vein occlusion • Eales' disease • Ocular ischaemic syndrome • Trauma • Intraocular tumours (choroidal melanoma) • Carotid-cavernous fistula.

Mnemonic: ABCD (medical treatment) — Atropine • Beta-blockers • Cidamex (oral acetazolamide) • Dexamethasone (topical steroid).

Read more: Baidya p.176–180, 437 • Namrata p.192–197 • Aravind 4.9, 4.15

G6 Pseudoexfoliation syndrome and pseudoexfoliation glaucoma

★ Kept once at JEH last year (with POAG) · Long case and short case

At a glance

- **Pseudoexfoliation (PXF):** white, flaky material on the pupillary margin, lens capsule, angle and cornea of older patients. **40%** develop glaucoma or ocular hypertension. **Pseudoexfoliation glaucoma (PXG)** is worse than primary open-angle glaucoma (POAG).
- Kept last year as “POAG with pseudoexfoliation glaucoma” (tasks: gonioscopy, applanation, Humphrey field); also on the short-case list.
- Examiners test: **four-point description, three zones, Sampaolesi’s line, true exfoliation, PXG vs POAG, PXF vs pigment dispersion, cataract-surgery precautions.**

Case format — history

Chief complaint: “Mr X, a 68-year-old man, has come for regular follow-up for glaucoma in the right eye” (or “...with diminishing distance vision in the right eye for 1 year”).

What to ask	Why it matters
History of present illness: defective vision — eye, onset, course	Nuclear sclerosis (commoner in PXF) or glaucoma
How glaucoma was found; sudden painful red eye	Usually silent, even with high intraocular pressure (IOP); a few have an acute IOP rise
Drops: names, number, compliance	One drug is usually not enough
Negative history: no coloured haloes, headache or vomiting	Rules out angle closure — primary, or from a lens pushed forward
No recurrent red, painful eye	Rules out uveitic glaucoma
No eye injury, herpes, glass blowing or infrared exposure	Rules out other causes of transillumination defects , and true exfoliation
No myopia since youth	Against pigment dispersion syndrome (PDS)
Past, drug and surgical: heart attack, stroke, hypertension, aortic aneurysm	PXF is systemic
Asthma; steroid use; sulpha allergy	No beta-blocker in asthma; house checklist (card G1)
Cataract, laser or glaucoma surgery	Intraocular lens (IOL) decentration; bleb (card G2)
Family: glaucoma or PXF	Genetic predisposition

Case format — examination

General and systemic (this case only)

- **Pulse, blood pressure, heart, nervous system, chest:** hypertension, old myocardial infarction or stroke, aortic aneurysm (PXF is systemic); asthma (no beta-blockers).

Ocular examination — RE | LE template

Columns: right eye (RE), left eye (LE).

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Nuclear sclerosis or glaucoma	6/24, pinhole 6/12
Extraocular movements, eyeball, lids	Deep-set eyes; relative anterior microphthalmos	Full; normal
Conjunctiva	Whitish powdery deposits; congestion with high IOP	Clear
Cornea	Flakes and pigment on endothelium (Krukenberg spindle); guttata; oedema	Flakes on endothelium
Anterior chamber (AC)	Van Herick (VH) grade — may be shallow; depth differs between eyes (zonular dialysis); flare	VH grade IV (ND, normal depth)
Iris	Flakes; transillumination defects at the pupillary margin ; atrophy; iridodonesis ; posterior synechiae	PXF at margin, no iris atrophy
Pupil (undilated)	Hallmark: white tufts on the margin; loss of ruff (moth-eaten margin); poor mydriasis	RTL 3 mm (round, reacting to light)

Structure	Look for in this case	How to record (example)
Lens (dilated)	Three zones ; nuclear sclerosis; phacodonesis ; subluxation	NS (nuclear sclerosis) grade 2; PXF in ALC (anterior lens capsule)
IOP (method, time)	Higher than in POAG; may exceed 40 mm Hg	Tn (tension) by GAT (Goldmann applanation tonometry) 18 mm Hg at 10 am
Gonioscopy	Open or occludable; blotchy pigment; Sampaolesi's line	Cross diagram (below)
Fundus	B-scan if cataract is mature; cupping with diffuse neuroretinal rim (NRR) loss; vein occlusion	CDR (cup-disc ratio) 0.8, diffuse NRR thinning
Fields	Confrontation	Full / defect ____

Special tests

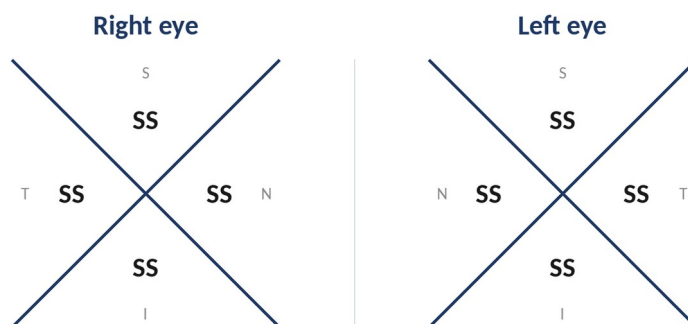
Slit lamp — the house four points

1. Undilated: pupillary margin (tufts, loss of ruff); retro-illuminate for transillumination defects.
2. Iridodonesis, phacodonesis; AC depth of both eyes.
3. Dilate after gonioscopy (note poor mydriasis): the three zones.
4. Describe: **colour** → **nature** → **location** → **with or without iris atrophy**.

Record: "White, flaky, dandruff-like material seen in the pupillary margin as tufts and on the anterior lens capsule as a disc-shaped deposit, with no associated iris atrophy — suggestive of pseudoexfoliation."

Gonioscopy (card G9)

1. Before dilating, each quadrant: the deepest structure seen (grading: card G9).
2. Blotchy pigment, dandruff-like deposits, **Sampaolesi's line** (on or in front of Schwalbe's line), occludable quadrants.



Deepest structure seen in each quadrant (SS = scleral spur) — open angles in all quadrants; note pigment and Sampaolesi's line beside the cross

Summary and complete diagnosis

Mr ___, a ___-year-old man on ___ drops for glaucoma in the ___ eye for ___ years, came for follow-up. On examination, visual acuity ___ / ___. The ___ eye showed PXF (four-point description), ___ mydriasis, ___ phacodonesis and ___ cataract. Tension by GAT ___ / ___ mm Hg at ___. Gonioscopy: ___ angles, Sampaolesi's line ___. CDR ___ with ___ rim thinning. Other eye: ___.

eye → disease → type / stage / grade → aetiology → complications or status → other eye

- "Right eye pseudoexfoliation glaucoma with immature senile cataract, intraocular pressure controlled on medication; left eye pseudoexfoliation syndrome."
- "Right eye pseudoexfoliation glaucoma with pseudophakia, intraocular pressure not controlled on two medications; left eye pseudoexfoliation syndrome with immature senile cataract."

Differential diagnosis

Condition	For	Against
Pigment dispersion syndrome, pigmentary glaucoma	Angle pigment; Krukenberg spindle; transillumination defects	Young, myopic, bilateral; mid-peripheral defects; uniform dark band
True exfoliation (capsular delamination)	Material on the anterior capsule	Clear scroll split from the capsule; heat or infrared; glaucoma infrequent
Primary amyloidosis	Whitish powdery deposits; glaucoma	Systemic; bilateral; fine deposits throughout the eye
POAG	Open angle; raised IOP; cupping	White material; usually one eye; Sampaolesi's line
Uveitic glaucoma	Flare; pigment; posterior synechiae	No AC cells or other signs of uveitis

Pseudoexfoliation glaucoma vs primary open-angle glaucoma

Feature	POAG	PXG
Age at onset	40–50 years	Over 60
Laterality	Bilateral	Usually unilateral
IOP	Above 21 mm Hg	Often above 40 mm Hg ; greater diurnal fluctuation
Disc	Focal or diffuse retinal nerve fibre layer (RNFL) defect	Usually diffuse ; more optic atrophy
Course	Good drug response; surgery sometimes	Faster ; poor drug response ; surgery often

Pseudoexfoliation vs pigment dispersion

Feature	Pigment dispersion	Pseudoexfoliation
Who	30–50 years; men; myopes; white race	After 50, mostly 7th decade; both sexes; Scandinavians; aortic aneurysm
Signs	Krukenberg spindle; deep AC; peripheral iris atrophy; Zentmayer's line	Target sign; weak zonules; poor dilatation; atrophy at the pupil edge
Gonioscopy	Heavily pigmented; uniform dark band	Blotchy pigment; Sampaoli's line

What you must know

Definition

- **Pseudoexfoliation syndrome (PXF, also PEX)**: an age-related systemic syndrome — gradual deposition of fibrillary, white, flaky material, mainly on the lens capsule, ciliary body, zonules, corneal endothelium, iris and pupillary margin, causing secondary glaucoma.
- **Glaucoma capsulare**: glaucoma in an eye with PXF; usually unilateral.
- **True exfoliation: lamellar delamination** of the anterior capsule into thin, clear scrolls that may float in the AC; from heat (**glassblower's cataract**) or infrared; prevented by goggles; also follows trauma, inflammation or age.
- **Who**: after 50, mostly in the 7th decade; men and women equally; commonest in Scandinavians; incidence **1.5–27%**; mostly bilateral.
- **Risk factors**: age above 70; Scandinavian; genetic predisposition (**LOXL1** gene); high blood homocysteine.
- **Systemic**: myocardial infarction, stroke, hypertension, aortic aneurysm; deposits in skin, heart, lungs, liver, kidneys, gall bladder, meninges.

Classification

Clinical classification by anterior lens capsule morphology:

Stage	Anterior capsule
Preclinical	Clinically invisible
Suspected PXF	Precapsular layer
Mini PXF	Focal defect, starting superonasally
Classic PXF	The three zones (below)

Three zones (target sign): translucent central disc, edges may curl (**absent in 20%**) · intermediate clear zone, rubbed clean by the moving iris · peripheral granular zone with radial striations (**seen only after dilatation**).

Pathogenesis

- **Material**: grey-white fibrillogranular matrix — fibrillin, α -elastin and laminin coated with glycosaminoglycans (hyaluronic acid) — from abnormal basement membranes of ageing epithelial cells.
- **Open angle**: material made in and deposited on the meshwork → swollen juxtacanalicular meshwork, narrowed Schlemm's canal; liberated iris pigment; trabecular cell dysfunction.
- **Angle closure**: lens-iris diaphragm forward on weak zonules; thick cataract; increased iris-lens adhesiveness.
- **Zonules and iris**: material invades zonule attachments and its proteolytic enzymes break them; sphincter atrophy → poor mydriasis, peripupillary defects; iris rubbing on the capsule sheds pigment.

Investigations

Test	Why	Expected finding
Diurnal variation of IOP	True IOP fall; target pressure	Wider swings than in POAG
Perimetry; RNFL imaging	Baseline and follow-up	Usually diffuse RNFL loss

Test	Why	Expected finding
Specular microscopy	Endothelium before surgery	Low cell count; altered size (polymegathism) and shape
Biometry (A-scan, keratometry); B-scan	IOL power; retina behind a mature cataract	Axial length; retina and disc
Ultrasound biomicroscopy (UBM)	Phacodonesis	Extent of zonular dialysis

Management

Treat **aggressively, with frequent follow-up**; allow for the wider diurnal swing in the target pressure.

1. **Medical:** the POAG classes — beta-blockers, alpha-adrenergic agonists, carbonic anhydrase inhibitors, prostaglandin analogues (strengths: card G1). **Monotherapy is usually insufficient: combine.**
2. **Laser:** argon or selective laser trabeculoplasty (ALT, SLT) — **higher success than in POAG, but shorter-lasting.**
3. **Surgery: trabeculectomy** if target IOP is not reached or glaucoma progresses (card G2); **combined with cataract surgery** for intractable IOP with cataract; **trabecular aspiration** removes intertrabecular debris.
4. **Other:** lower homocysteine if high; antioxidants.
5. **Follow-up: yearly** without glaucoma, **3-monthly** with glaucoma.

Cataract surgery. First check endothelium, phacodonesis, AC depth asymmetry (zonular dialysis: confirm by UBM), dilatation, synechiae, subluxation.

Problem	Precaution
Poorly dilating pupil	Atropine; stretch the pupil; iris or Kuglen hooks; expansion ring; mini-sphincterotomies; sector iridectomy
Endotheliopathy	Healon 5 (visco-adaptive) or Viscoat (visco-dispersive); chondroitin sulphate; Arshinoff soft-shell technique
Weak zonules	Chop, or prolapse the nucleus out of the bag; capsular tension ring (CTR) ; 3-piece IOL; or extracapsular surgery
Inflammation	Aggressive, longer steroids; heparin-surface-modified IOL
Pressure spike	Postoperative acetazolamide (Diamox)
Capsular phimosis	Nd:YAG (neodymium-doped yttrium aluminium garnet) laser relaxing incisions at the 4 cardinal positions

Complications

- **Glaucoma:** open-angle, angle-closure, acute pressure rises.
- **Lens:** nuclear cataract, phacodonesis, zonular dialysis, subluxation.
- **Iris:** posterior synechiae, uveitis, sphincter degeneration, poor mydriasis.
- **Cornea, retina:** early endothelial decompensation; central retinal vein occlusion, retinal detachment.
- **Surgical:** vitreous loss, IOL decentration, corneal decompensation, inflammation, IOP rise.

Prognosis and counselling

- Worse than POAG: higher IOP, faster progression, poor drug response.
- Counsel: lifelong drops; riskier cataract surgery; yearly review of the other eye; a heart check.

Recent advances

- **LOXL1 gene:** an exon 1 polymorphism is associated with exfoliation syndrome. Nothing else in your practical books.

Viva questions

Why did you ask / examine...?

Q Why did you ask about heart attack, stroke and blood pressure?

PXF is systemic: linked with myocardial infarction, cerebrovascular events, hypertension and aortic aneurysm.

Q Why did you ask about glass blowing?

To exclude true exfoliation — capsular delamination from heat or infrared.

Q Why did you look at the pupil before dilating?

The hallmark flecks and loss of ruff are on the pupillary margin; poor dilatation is itself a sign.

Q Why did you dilate, and look for phacodonesis?

The peripheral zone appears only after dilatation. Phacodonesis and asymmetric AC depth mean zonular dialysis, which changes cataract surgery.

Diagnosis

Q What is pseudoexfoliation syndrome?

Age-related systemic deposition of fibrillary, white, flaky material on the lens capsule, zonules, ciliary body, cornea, iris and pupillary margin, causing secondary glaucoma.

Q How is true exfoliation different?

The capsule itself delaminates into clear scrolls after heat or infrared exposure (glassblower's cataract); glaucoma is infrequent.

Q What is glaucoma capsulare? How often does PXF cause glaucoma?

Glaucoma in an eye with PXF; usually unilateral. Glaucoma or ocular hypertension in 40%; PXG in up to 50% of eyes.

Q What are the hallmark and the most consistent signs?

Hallmark: flecks on the pupillary margin. Most consistent: the whitish powdery ring on the anterior capsule.

Q Describe the lens deposit.

Target sign: central disc (absent in 20%), clear zone rubbed by the iris, peripheral granular zone after dilatation.

Q What is Sampaolesi's line? Is it specific?

Dark, dense, scalloped pigment on or anterior to Schwalbe's line. Not specific: also in pigment dispersion and chronic inflammation.

Q How does PXF raise IOP?

Open angle: material and liberated pigment clog the meshwork; trabecular cells fail. Angle closure (occludable in 9–18%): lens forward on weak zonules, thick cataract, iris–lens adhesion.

Q How does PXG differ from POAG?

Older, usually unilateral, higher and more fluctuating IOP, diffuse rim damage, faster progression, poor drug response.

Q How do you tell PXF from pigment dispersion?

PDS: myopic men aged 30–50, posterior iris bowing, mid-peripheral defects, uniform dark band. PXF: after 50, target sign, blotchy pigment, Sampaolesi's line.

Q Why are the zonules weak?

Material invades their attachments and its proteolytic enzymes digest them; PXF zonules are modified fibres.

Investigations

Q What does specular microscopy show?

Low endothelial cell density; altered cell size and shape.

Q When will you order ultrasound biomicroscopy?

Phacodonesis before cataract surgery: it confirms the extent of zonular dialysis.

Management and surgery

Q How will you treat this glaucoma?

Aggressively: combination drops, laser trabeculoplasty, then trabeculectomy if target IOP is not reached; combined surgery with cataract.

Q Is laser trabeculoplasty useful?

Yes: more successful than in POAG, but shorter-lasting.

Q What precautions will you take for cataract surgery?

Pupil: hooks or ring. Endothelium: dispersive viscoelastic. Zonules: chop, capsular tension ring, 3-piece IOL. After: longer steroids, acetazolamide (Diamox).

Complications

Q What complications do you expect with cataract surgery?

Zonular dialysis, vitreous loss, IOL decentration or late prolapse into the posterior segment, corneal decompensation, inflammation, IOP rise.

Q What are the posterior-segment complications of PXF?

Central retinal vein occlusion; retinal detachment.

Recent advances

Q Which gene is associated with exfoliation syndrome?

LOXL1 — a polymorphism in exon 1.

Examiner traps

- Confusing pseudoexfoliation with true exfoliation (capsular delamination from heat or infrared).
- Dilating before showing the pupillary margin, or calling the lens clear without dilating.
- Calling Sampaolesi's line specific, or swapping the angles: **blotchy = PXF, uniform dark band = pigment dispersion**.
- Treating PXG like POAG: one drug, one IOP reading, no phasing.
- Offering cataract surgery without checking phacodonesis, AC depth of both eyes, dilatation and endothelium; forgetting the other eye.

Short-case version

Spot diagnosis: “White, flaky, dandruff-like material on the pupillary margin and anterior lens capsule — suggestive of pseudoexfoliation.”

Focused examination, in order:

1. Undilated pupillary margin: white tufts, loss of ruff; four-point description.
2. Iris transillumination (retro-illumination): defects at the pupillary margin.
3. Phacodonesis, iridodonesis; AC depth of both eyes.
4. Gonioscopy, before dilating: blotchy pigment, Sampaolesi’s line, angle width.
5. After dilatation: poor mydriasis; three zones on the anterior capsule.

3 signs to demonstrate: white flecks on the pupillary margin (hallmark) · three-zone target sign · Sampaolesi’s line.

Say-it script

SAY IT

Mr X, a sixty-eight-year-old man, has come for regular follow-up of glaucoma in the right eye, on two eye drops for three years, with gradual, painless diminution of distance vision in that eye for one year. There is no history of coloured haloes, recurrent red eye, trauma or glass blowing.

Visual acuity is 6/24 in the right eye, 6/12 with pinhole, and 6/9 in the left. The right eye shows white, flaky, dandruff-like material in the pupillary margin as tufts and on the anterior lens capsule as a disc-shaped deposit, with no associated iris atrophy, suggestive of pseudoexfoliation. The pupil dilates poorly, without phacodonesis, and there is an immature senile cataract. Intraocular pressure by Goldmann applanation is 18 millimetres of mercury in the right eye and 15 in the left at ten in the morning. Gonioscopy shows open angles, with blotchy pigmentation and Sampaolesi’s line on the right. The right disc has a cup–disc ratio of 0.8 with diffuse rim thinning. The left eye shows flakes at the pupillary margin.

My diagnosis is right eye pseudoexfoliation glaucoma with immature senile cataract, intraocular pressure controlled on medication; left eye pseudoexfoliation syndrome.

I would do diurnal variation of intraocular pressure, perimetry and specular microscopy, continue combination therapy, and plan cataract surgery with precautions for poor dilatation and weak zonules, keeping a capsular tension ring ready.

Quick recall

- **Four points:** colour (white) · nature (flaky, dandruff-like) · location (margin tufts; disc on the capsule) · with or without iris atrophy.
- **Hallmark:** flecks on the pupillary margin. **Most consistent:** whitish ring on the anterior capsule.
- **Three zones:** central disc (absent in 20%) · clear zone · peripheral granular zone (only after dilatation).
- **Sampaolesi’s line:** on or in front of Schwalbe’s line; not specific.
- **40%** develop glaucoma or ocular hypertension; occludable angle in **9–18%**.
- **PXG:** over 60, usually unilateral, IOP often above 40 mm Hg, wide diurnal swings, poor drug response.

Mnemonic: PIGMENT — causes of trabecular pigmentation: Pseudoexfoliation and Pigment dispersion syndrome · Iritis · Glaucoma (after angle closure) · Melanosis of the angle (oculodermal melanosis) · Endocrine (diabetes, Addison’s) · Naevus (Cogan–Reese syndrome) · Trauma.

Read more: Baidya p.167–169 · Namrata p.203–207 · Aravind 4.11

G7 CRVO with POAG, including hemi-CRVO with POAG

★ Kept twice at JEH last year (CRVO with POAG as a glaucoma case; hemi-CRVO with POAG as a fundus case) · Short case

The instruction you may get

- “Examine the fundus of this glaucoma patient and draw it.”

Spot

- **Flame-shaped and dot-blot haemorrhages** radiating from the disc, with **dilated, tortuous veins**: four quadrants (“**blood and thunder**”) in central retinal vein occlusion (CRVO).
- **Hemiretinal vein occlusion** (*Kanski — not in your practical books*): regarded by some as a CRVO variant, ischaemic or non-ischaemic, occluding the **superior or inferior branch** of the central retinal vein: **hemispheric** (a major branch at or near the disc) or **hemicentral** (one trunk of a dual-trunked central retinal vein). Branch-occlusion picture in that **hemisphere**. Neovascularisation of the iris: commoner than in branch, rarer than in central occlusion. Extensive ischaemia → manage as **ischaemic CRVO**.
- **Glaucomatous cupping**, same or fellow eye.
- Old eye: **optociliary shunts, sheathed veins, laser scars, epiretinal membrane**.

Focused examination

1. **Undilated, both eyes: relative afferent pupillary defect (RAPD)**: unreliable with **glaucomatous discs** or miotic pupils.
2. **Pupillary margin: neovascularisation of the iris (NVI)**, in clock hours.
3. **Intraocular pressure (IOP)**: over 22 mm Hg needs a cause: open-angle glaucoma, neovascular glaucoma (NVG), angle closure.
4. **Undilated gonioscopy: neovascularisation of the angle (NVA)**, without NVI in 12%; occludable angle.
5. **Lens**: nuclear sclerosis grade.
6. **Dilate. Distant direct ophthalmoscopy**: red glow; **media**.
7. **Disc**: size, shape, colour, margins, **cup-disc ratio**, rim; **glaucomatous cupping**; swelling or pallor; shunts or **new vessels on the disc (NVD)**.
8. **Vessels**: dilated, tortuous veins, all quadrants or one hemisphere; sheathing.
9. **Background**: each lesion by colour, shape, size in disc diameters, margins, number, location → “suggestive of...”: flame (superficial) and dot-blot (deep) haemorrhages, cotton-wool spots.
10. **Macula**: foveal reflex, oedema, epiretinal membrane.
11. **Periphery, indirect ophthalmoscope: new vessels elsewhere (NVE)**.
12. **“Confirmed with a 90 D (dioptre) lens”**: tells shunts from NVD.
13. **Draw both fundi**: red haemorrhages, blue veins, yellow sclerosed vessels, black laser scars, green epiretinal membrane.

Say-it description

SAY IT

Mr X, a 65-year-old hypertensive man with primary open-angle glaucoma on drops, has sudden painless defective vision in the right eye for two months.

Vision: right eye 6/36, no better with pinhole; left eye 6/9. Non-contact tonometry: 18 and 16 millimetres of mercury.

Both eyes: conjunctiva, cornea, anterior chamber and iris normal, no new vessels; no relative afferent pupillary defect; grade 2 nuclear sclerosis; undilated gonioscopy: open angles, no new vessels.

Right fundus: good red glow, clear media. Disc vertically oval, pink, well-defined margins; cup-disc ratio 0.7, inferior rim thinning, nasalisation, suggestive of glaucomatous cupping. Superior veins dilated, tortuous. Multiple red flame-shaped lesions, a quarter to half a disc diameter, ill-defined margins, with red dots and blots, radiating from the disc into the superior hemisphere only, suggestive of superficial and deep retinal haemorrhages; no cotton-wool spots. Foveal reflex absent, retina thickened, suggestive of macular oedema. Indirect ophthalmoscopy: no peripheral new vessels. Confirmed with a 90 dioptre lens.

Left fundus: cup-disc ratio 0.8, inferior rim thinning, suggestive of glaucomatous cupping.

Right eye clinically **non-ischaemic hemi-central retinal vein occlusion** involving the superior hemisphere, with macular oedema, in a known case of primary open-angle glaucoma in both eyes, intraocular pressure under control on medical treatment; both eyes grade 2 nuclear sclerosis.

Differentials

Condition	Distinguishing point
Branch retinal vein occlusion	Collaterals cross the horizontal raphe (central: at the disc); more retinal than iris new vessels
Ocular ischaemic syndrome	Veins not tortuous ; mid-peripheral haemorrhages; normal disc; carotid stenosis

Condition	Distinguishing point
Diabetic retinopathy	Always bilateral; beaded veins; hard exudates

Must know

- **Types (Central Vein Occlusion Study, CVOS):** ischaemic; non-ischaemic, about **75% (34% turn ischaemic by 3 years)**; **indeterminate** (haemorrhage hides perfusion).

Feature	Non-ischaemic	Ischaemic
Vision	Moderate to severe loss	Counting fingers or worse
RAPD	Absent or mild	> 0.7 log unit
Fundus fluorescein angiography (FFA) non-perfusion	< 10 disc areas	> 10 disc areas
Electroretinogram (ERG)	Normal or mildly depressed	b-wave < 60% of normal; b-to-a ratio < 1

- **Grid laser:** helps branch occlusion, not central (diffuse leakage involving the fovea).
- **Glaucoma both ways:** open-angle glaucoma heads the ocular associations. CRVO raises IOP by NVG, **steroid-induced glaucoma** or angle closure (cause or effect).
- **“100-day glaucoma”:** NVI usually at **2–4 months**; ischaemic: **NVI 49%, NVG 29%** within **6 months**; **one-third** with NVI get NVG. Significant: **NVI over 2 clock hours**, or **any NVA**.
- **Investigations:** FFA once haemorrhages clear (about **3 months**); optical coherence tomography (OCT); blood pressure, fasting blood glucose, lipid profile; hypercoagulable screen if under 60.
- **Macular oedema:** intravitreal **anti-vascular endothelial growth factor (anti-VEGF)** first (**ranibizumab, aflibercept, bevacizumab**); steroids (**triamcinolone, dexamethasone implant**) cause cataract and **raise IOP**.
- **New vessels:** **prompt panretinal photocoagulation (PRP)**; **not prophylactic** unless ischaemic with follow-up impossible; anti-VEGF only temporises.
- **IOP and follow-up:** antiglaucoma drugs, then trabeculectomy or drainage valve. Review **monthly for 6 months**, then tapered (CVOS), with undilated gonioscopy each visit. See G1 and G5.

Viva questions

Q Why does glaucoma predispose to CRVO?

Retinal venous pressure must exceed IOP, so raised IOP causes stasis (Virchow's triad); the backward-bowed lamina cribrosa and nasalisation compress the vein.

Q What will you look for at every follow-up visit?

Vision, RAPD (conversion to ischaemic), IOP, undilated NVI and NVA, NVD, NVE, macular oedema (OCT), fellow eye.

Q Why is RAPD unreliable here?

Needs normal discs, pupils and fellow eye; ERG does not (b-wave: **80%** sensitivity and specificity).

Q Why call it non-ischaemic?

Vision better than counting fingers, no RAPD, no cotton-wool spots; FFA confirms (under 10 disc areas).

Q Why not prophylactic PRP?

CVOS: no prevention of NVI or NVA; later PRP regressed them better in untreated eyes (**56%** versus **22%**).

Q Why is NVI commoner than NVD?

Retina too ischaemic to form new vessels; anterior segment capillaries stay viable.

Q Shunt or NVD?

Shunts drain retinal blood to the choroid (good compensation, old occlusion); NVD are finer, forming a network.

Q Ischaemic index?

Non-perfused ÷ total retinal area; over 50% (about 10 disc areas): significant neovascular risk.

Q Commonest cause of visual loss?

Macular oedema.

Q Commonest field defect?

Central scotoma.

Q Why gonioscopy in the fellow eye?

An occludable angle, as the occlusion may follow angle closure.

Read more: Baidya p.264–273 · Namrata p.216–222 · Aravind 6.6, 4.8 · Kanski p.521

G8 Applanation tonometry — perform and record

★ Done in 8 of the 12 glaucoma cases at JEH last year · Task

Indications

- Intraocular pressure (IOP) in **every glaucoma case**, both eyes.
- Goldmann applanation tonometry (GAT): **reliable, reproducible, not influenced by scleral rigidity**.
- Other tonometers:
 - cornea scarred, irregular or oedematous, bandage contact lens, nystagmus → **Tonopen** (scarring: also **non-contact**, **Mackay-Marg**, **pneumatic**)
 - bed-bound or anaesthetised → **Perkins** (hand-held)
 - community screening → **non-contact tonometer** (no anaesthetic)

Instrument check

- **Calibration** checked each session (manufacturer's **calibration arm**).
- **Dial at 1** (10 mm Hg), never 0: **microvibrations** can cause **corneal erosions**.
- **Astigmatism over 3 dioptres**: rotate the prism to align the **minus-cylinder axis** with the **red line** on the housing, or average two readings, prism turned **90°** for the second.

Steps

Observed-station checklist: consent, patient instruction, tonometer checked, topical drops applied, handling, (Schiotz) conversion chart referred to.

1. **Introduce yourself**, explain the test, take **consent**.
2. **Instruct**: "Chin on the rest, forehead against the headband. Look straight ahead. Keep both eyes open. Do not squeeze. Breathe normally."
3. **Check the tonometer**: calibration, prism clean, **dial at 1**, prism rotation.
4. **Disinfect** the prism tip: **70% isopropyl alcohol**, **3% hydrogen peroxide** or **diluted sodium hypochlorite (1:10)**.
5. Instil a **topical anaesthetic drop** and **fluorescein** (strip or **0.25% dye**).
6. Slit lamp: **cobalt blue filter**, **maximal illumination**, light and microscope **60°** apart.
7. Separate the lids gently if needed: **no pressure on the globe**.
8. Advance until the prism **just touches the corneal apex**: two **green semicircular mires**, above and below the horizontal midline.
9. Turn the dial until the mires are **equal** and their **inner edges just touch**.
10. **Dial reading × 10 = IOP in mm Hg** (1.8 → 18 mm Hg). Repeat for the other eye; avoid quick repeats.
11. After use, **disinfect the prism again**.

How to record the result

IOP (GAT) right eye (RE) 18 mm Hg, left eye (LE) 16 mm Hg at 10:30 am: **eye, method, value, time**. Case sheet: **Tn (tension) by GAT** row; central corneal thickness (CCT) goes under investigations.

- **Time** (diurnal variation): usually highest in the morning, lowest in the evening; swings **3–6 mm Hg** in normal eyes, **10 mm Hg or more** in glaucomatous eyes; peaks **3–4 hours after the cortisol peak**.

Normal values and interpretation

- Normal IOP **10–21 mm Hg**. **Why 21**: mean **15.5 ± 2.57**; mean + 2 standard deviations ≈ **20.5** (95% of a Gaussian curve lies within ± 2). The curve is **skewed to the right**, so 21 is only approximate.
- Basis: **Imbert-Fick law** ($P = F/A$). At **3.06 mm**, corneal rigidity and the tear meniscus's **capillary attraction** cancel out.
- **CCT** (mean **520 μm**): thick **over-reads**, thin **under-reads**, about **0.7 mm Hg per 10 μm**. **Oedema** reads falsely **low**.

Common errors

- **Mires too thick / too thin** (inappropriate fluorescein pattern) → falsely **high / low**.
- **Unequal semicircles** (improper vertical alignment) → falsely **high**.
- **Squeezing**, or pressure on lids or globe → IOP **raised** (orbicularis tone).
- **Breath-holding or straining** (Valsalva manoeuvre) → IOP **raised** (episcleral venous pressure).
- **Astigmatism** → with-the-rule **under-reads**, against-the-rule **over-reads**, about **1 mm Hg per 4 dioptres**.
- **Thick, thin, oedematous, scarred or irregular cornea**, or after corneal refractive surgery → **inaccurate** (see CCT).
- **Repeated readings** in quick succession → slight **fall** (massaging effect).

Viva questions

Q State the Imbert-Fick law.

Pressure in an ideal dry, thin-walled sphere = force to flatten its surface ÷ area flattened ($P = F/A$).

Q Why 3.06 mm?

Corneal rigidity cancels tear-meniscus capillary attraction, and $0.1 \text{ g} = 1 \text{ mm Hg}$ (hence dial $\times 10$).

Q Why fluorescein and cobalt blue light?

Stained tears fluoresce, showing the **tear meniscus** at the prism–cornea edge as the two semicircles.

Q CCT 600 μm , GAT 26 mm Hg. Comment.

Over-reads by about $80/10 \times 0.7 \approx 5.6 \text{ mm Hg}$: corrected IOP $\approx 20 \text{ mm Hg}$.

Q Why is applanation better than Schiøtz?

Not influenced by scleral rigidity; Schiøtz under-reads in high myopia and after ocular surgery, over-reads in high hypermetropia.

Q Which infections can a tonometer spread?

Adenovirus (epidemic keratoconjunctivitis), **herpes simplex virus type 1**, **human immunodeficiency virus**, **hepatitis B**.

Q How is Schiøtz tonometry done?

Supine, cornea anaesthetised; zero checked on the metal plate; footplate on cornea without pressing the globe; scale reading → **conversion chart**. Weights 5.5, 7.5, 10, 15 g.

Q Which tonometer ignores corneal thickness?

Dynamic contour tonometer (PASCAL): contour-matching surface, digital display, no fluorescein.

Read more: Baidya p.738–740 • Aravind 4.1

G9 Gonioscopy — perform, grade and draw

★ Done in about 8 of the 12 glaucoma cases at JEH last year - Task

Indications

- **Every glaucoma case, both eyes:** open or closed angle; **early narrow angles**; extent of closure; **iridotomy patency**.
- **Secondary glaucomas:** angle recession, new vessels, **pseudoexfoliation**, pigment. Also tumour, foreign body, blood, **Kayser-Fleischer ring**.
- Therapeutic: indentation can **break an acute angle-closure attack**.

Instrument check

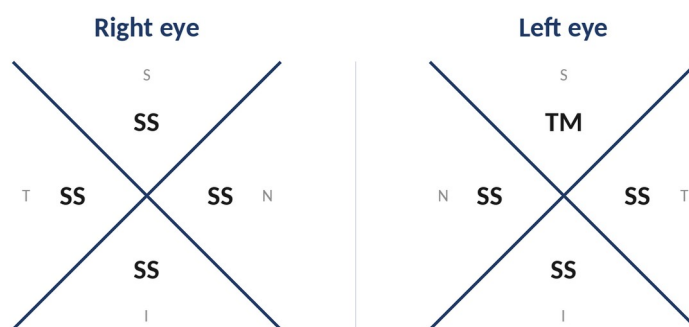
- **Indirect** (slit lamp): mirror shows the **opposite** angle.
 - **Goldmann** three-mirror: angle mirror **smallest, dome-shaped**; contact **12 mm**, steeper than cornea → **viscous coupling agent**; steadies the globe.
 - **Four-mirror** (**Zeiss, Posner, Sussman**): contact **9 mm**, **no coupling agent**; whole angle at once; **indentation**.
- **Direct** (Koepe): supine, usually under general anaesthesia (infantile glaucoma).

Steps

1. **First:** Van Herick, then **intraocular pressure**. Corneal oedema (acute attack): **defer**.
2. **Consent:** introduce yourself, explain, take consent.
3. **Instruction:** “Numbing drops, then a lens on your eye; keep still, look where I ask.”
4. **Topical anaesthetic drops.**
5. **Disinfect** the lens.
6. **Dim the room.** Slit beam **2 mm** wide, perpendicular to the mirror.
7. Place the lens gently; **grade before any pressure**.
8. Rotate **360°**. Mirror at 12 o'clock: **inferior** angle. Narrow angle: patient **looks towards the mirror**.
9. **Iris root forwards:** **ciliary body band (CBB)** → **scleral spur (SS)**, white → **trabecular meshwork (TM)**, posterior **pigmented** (main outflow), anterior non-pigmented → **Schwalbe's line (SL)**.
10. **Indentation** (four-mirror): press gently. **Appositional** closure opens; **peripheral anterior synechiae (PAS)** stay closed.
11. **Note:** iris contour, TM pigment, PAS extent, new vessels, angle recession, pseudoexfoliation, blood.

How to record the result

One cross per eye, right eye (RE) and left eye (LE); each quadrant: **deepest structure seen** (SS, CBB, TM, SL or “Closed”).



Deepest structure seen in each quadrant — RE open angles in all quadrants; LE open, superior quadrant narrower (pigmented TM only); neither eye occludable

Written: “RE SS all round: open angles. LE superior pigmented TM only, rest SS. No PAS or new vessels.” Closed quadrant: “appositional” or “PAS”.

Your department writes a grade in each quadrant “by modified Shaffer’s grading” (III in all four on its primary open-angle glaucoma sheet); confirm with your seniors which structure each grade stands for, and write the structure too.

Normal values and interpretation

Shaffer	Width	Structures seen	Closure
4 wide open	35–40°	Up to ciliary body	Impossible
3 open	25–35°	Up to ciliary body	Impossible
2 moderately narrow	20°	Pigmented TM	Possible, unlikely
1 extremely narrow	10°	Only SL ± non-pigmented TM	Possible

Shaffer	Width	Structures seen	Closure
S slit	< 10°	None; no obvious contact	Parts look closed
0 closed	—	Iridocorneal contact	Complete

Occludable angle: pigmented TM not seen (Shaffer 1 or 0) without indentation or manipulation in ≥ 3 quadrants.

System	Grades
Scheie	Wide open all seen · narrow 1 iris root hard to see · 2 CBB obscured · 3 posterior TM obscured · 4 only SL
Spaeth	Four parameters: insertion A anterior to TM, B at TM, C at SS, D deep to SS, E into ciliary body · width 10°/20°/30°/40° · contour S steep, R regular, Q queer (deeply concave) · TM pigment 0–4

Van Herick (peripheral depth ÷ corneal thickness): $4 \geq 1$ (normal) · $3 \frac{1}{4}$ – $\frac{1}{2}$ · $2 \frac{1}{4}$ · $1 < \frac{1}{4}$, **potentially occludable**.

Common errors

- **Pressure** while grading opens **appositional** closure.
- Charting the mirror's quadrant, not the **opposite** one.
- Pigmented SL or **Sampaolesi's line** taken for pigmented TM (grade 1 called 2).
- Closed angle called synechial without **indentation**.
- Iris processes taken for PAS.

Viva questions

Q Why a gonioscope, and its principle?

Total internal reflection at the tear–air interface hides the angle; the lens replaces that interface with one of **higher refractive index**. An indirect mirror sends rays out **below the critical angle**.

Q Direct versus indirect?

Direct: erect view, simple orientation, sees over a convex iris; supine, slow. Indirect: quick, allows compression; orientation confusing.

Q Indentation (dynamic) gonioscopy?

Pressing a four-mirror lens: **appositional** closure opens, **synechial** stays; shows PAS extent.

Q What is the scleral spur?

Posterior lip of the scleral sulcus; attachment of the **longitudinal ciliary muscle**.

Q Iris processes versus PAS?

Processes **lacy, fenestrated**, angle seen between; PAS **solid**, hide it.

Q New versus normal angle vessels?

New: **fine, cross the SS, arborise** on TM. Normal: broad, do neither.

Q Sampaolesi's line versus pigmented TM?

Salt-and-pepper, dark, granular, discontinuous versus **"brown sugar"**, fine, continuous.

Q Gonioscopy after trauma?

Angle recession, trabecular dialysis, cyclodialysis, foreign body, blood.

Mnemonic: angle structures: "I CAN SEE TILL SCHWALBE'S LINE" · Goldmann three-mirror: "The smallest the lens, the more peripheral the view".

Read more: Baidya p.148–150, 741–742 · Namrata p.180–181, 183 · Aravind 4.2

G10 Humphrey visual field in glaucoma — reading the printout

★ Done in 3 of the 12 glaucoma cases at JEH last year · Chart

What it is

- **Static automated threshold perimetry**: a still light, brightened until seen; **threshold** = the intensity seen **50%** of the time. **Gold standard** for glaucoma diagnosis and management.
- **Goldmann size III white stimulus, 31.5 apostilb (asb)** background. **Decibels (dB)**: **0 dB = 10,000 asb, 40 dB = 1 asb**; higher = more sensitive.
- **24-2** (54 points) and **30-2** (76): **6° grid**; **10-2** (68): **2° grid**, advanced glaucoma.
- **Swedish interactive threshold algorithm (SITA)**: half the time of full threshold, little or no accuracy lost.

Reading order

Reliability first; name the defect on the **pattern deviation** plot.

1. **Zone 1, patient data** (top strip): **eye, age** (age-matched normals).
2. **Zone 2, test conditions** (top centre, right): programme, strategy, **pupil (<2 mm: dilate)**, acuity, **correcting lens**.
3. **Zone 3, reliability** (top left): **fixation losses <20%** (fraction: $8/19 = 42\%$, unreliable); unreliable if **false positives >15%** (**white scotomas**) or **false negatives >33%** (**clover-leaf**; may mean advanced glaucoma); **fovea** should fit acuity.
4. **Zone 4, grey scale** (top right): picture of raw values; no statistics.
5. **Zone 5, raw data** (top left): threshold in dB; **0 = absolute scotoma**.
6. **Zone 6, deviation plots** (lower half: numbers, then probability (p) symbols): **total deviation (TD)** (left), patient minus age-matched normal: overall and focal loss; **pattern deviation (PD)** (right), TD shifted so its **7th best point** reads 0: overall depression removed, **localised defects** left.
7. **Zone 7, glaucoma hemifield test (GHT)** (right): **5 sectors above versus mirror images below: outside normal limits ($p < 1\%$), borderline ($p < 3\%$)**, within normal limits, general reduction, abnormally high sensitivity.
8. **Zone 8, global indices** (below GHT): **mean deviation (MD)**, overall loss (normal typically within 2 dB of 0; more negative = worse); **pattern standard deviation (PSD)**, focal loss; **visual field index (VFI)**, % of normal, from PD (unaffected by cataract).
9. **Conclude**: defect, Anderson criteria, stage, **matching disc and retinal nerve fibre layer (RNFL) damage** (superior defect: inferior pole).

Findings to name

From **Bjerrum's area (10–25° from fixation)**, respecting the **horizontal meridian**, in order:

1. **Generalised depression**: earliest, not diagnostic.
2. **Baring of the blind spot**: early, non-specific.
3. **Paracentral scotoma**: **earliest clinically significant**; mostly **superonasal**.
4. **Seidel's scotoma**: joins the blind spot; **sickle-shaped**.
5. **Arcuate (Bjerrum's) scotoma**: blind spot to **horizontal raphe**.
6. **Ring (double arcuate) scotoma**.
7. **Nasal (Roenne's) step**: right-angled, at the horizontal midline.
8. **Tubular vision and temporal island**.

Anderson and Patella criteria (at least 2 consecutive reliable fields)

- **3 or more contiguous non-edge points** on the 30-2 PD plot, all $p < 5\%$, one $p < 1\%$
- **PSD or corrected PSD (CPSD) $p < 5\%$**
- **GHT outside normal limits**

Grading or classification

All stages: **CPSD $p < 5\%$, GHT outside normal limits**.

Parameter	Early	Moderate	Severe
MD	-6 dB or better	-6 to -12 dB	-12 dB or worse
PD points $p < 5\%$ / $p < 1\%$	<18 / <10	<37 / <20	>37 / >20
Central 5° points <15 dB	None	One hemifield	Both hemifields
Central 10°	Spared	Spared	Involved

What each finding leads to (management)

- **Unreliable**: do not diagnose; re-instruct, **repeat** (criteria need **2 reliable fields**).
- **Generalised depression, PD normal**: cataract, miosis or uncorrected refraction; correct it, **dilate if pupil <2 mm**, repeat.

- **Criteria met, matching the disc: glaucomatous field defect;** stage it.
- **Central 10° involved:** serial **10-2**; **size V** for the macular split.
- **Worse on serial fields** (falling MD or VFI; deeper, wider or new scotoma; **trend or glaucoma progression analysis**): **progression**; review treatment.
- **Does not fit** (vertical meridian, no disc match): artefact, retinal, neurological or other optic nerve disease.

Viva questions

Q Why PD, not TD?

Removes generalised depression (cataract, small pupil), unmasking localised defects.

Q Effect of cataract?

TD and MD depressed; PD near normal; PSD low; VFI unaffected.

Q What does $p < 2\%$ mean?

Under 2% of normals are this low there; significance, not depth.

Q Glaucomatous versus neurological defects?

Glaucomatous: horizontal meridian, relative, incongruous. Neurological: vertical, absolute, mostly congruous.

Q Other causes of an arcuate scotoma?

Myopic peripapillary atrophy, disc drusen, ischaemic optic neuropathy, pituitary adenoma.

Q Point criteria for progression?

Reproducible loss > 7 dB within a scotoma, > 9 dB beside it, > 11 dB at a normal point.

Q False positives 20%?

Unreliable (over 15%): trigger-happy, white scotomas. Repeat.

Q Perimetry for earlier detection?

Short-wavelength automated (blue-on-yellow) and frequency-doubling technology.

Read more: Baidya p.163, 181–194 · Aravind 4.4

G2 POAG with trabeculectomy — bleb assessment and the failed bleb

On the department checklist (not kept last year) · Long case

At a glance

- **Primary open-angle glaucoma (POAG)** after **trabeculectomy**; the bleb has **failed** and the intraocular pressure (IOP) is above target.
- Checklist case: lasting signs (bleb, surgical iridectomy, cupped disc), long surgical viva.
- Tested: **bleb in house order**, Seidel's test, iridectomy patency, **internal ostium**, antimetabolites, the shallow chamber.

Case format — history

Chief complaint: "Mr X, a 56-year-old man, came with complaints of defective vision in the right eye (RE) for the past 1 year and the left eye (LE) for the past 6 months; the right eye was operated for raised pressure 6 months ago."

What to ask	Why it matters
History of present illness: which eye, when	Early and late complications differ
Why operated	Indication: failed medical therapy, poor compliance, progression
Mitomycin C (MMC) or 5-fluorouracil (5-FU) used?	More hypotony, bleb leak, bleb infection; more with MMC
After surgery: eye pressed, laser, injections, revision	What was tried for a failing bleb
Present drops	Restarting drops: a step for a failing filter
Negative history: no redness, watering, pain	Rules out bleb infection (earliest sign: hyperaemia around the bleb) and endophthalmitis
No sudden loss of vision	Rules out snuff-out , central retinal vein occlusion, acute IOP rise
Past, drug and surgical: years on drops before surgery	Long-term drops: more fibroblasts, fewer goblet cells; over 3 years → antimetabolite
Conjunctival or cataract surgery, keratoplasty, scleral buckle, uveitis	Poor prognosis; antimetabolite
Fellow eye	Tenon's cyst there is a risk factor; failed 5-FU surgery there → MMC
Asthma, heart failure, bradycardia; sulpha allergy; aspirin	Restarted drops: no beta-blocker; no carbonic anhydrase inhibitor in sulpha allergy. Aspirin off 5 days before surgery

Case format — examination

General and systemic (this case only)

- **Chest and pulse:** wheeze or bradycardia → no beta-blocker when drops restart.

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity (pinhole)	Fall since surgery: cataract, snuff-out	6/60, pinhole 6/12
Lids	Lid retraction; punctal occlusion (5-FU)	Normal
Conjunctiva	Bleb in slide order (Special tests); hyperaemia around it	"Flat, opaque bleb, 11 to 2 o'clock superiorly above the limbus; dilated tortuous vessels; no microcysts; subconjunctival fibrosis; no sutures — suggestive of a failed bleb"
Cornea	Dellen; 5-FU or MMC toxicity	Clear
Anterior chamber (AC)	Central and peripheral depth; cells; hyphaema	Van Herick (VH) grade IV (ND, normal depth)
Iris	Peripheral iridectomy (PI): margin, site, surrounding iris, patency	"Sharp margin; 2 mm, 12 o'clock; iris normal; retro-illumination positive — patent surgical iridectomy"
Pupil	Relative afferent pupillary defect (RAPD)	Grade 3 RAPD
Lens	Cataract	Nuclear sclerosis grade 2
IOP	Goldmann applanation tonometry (GAT) before gonioscopy; time; target	Tn (tension) by GAT 26 mm Hg at 10 am

Structure	Look for in this case	How to record (example)
Gonioscopy	Internal ostium patent or blocked; PI; peripheral anterior synechiae (PAS)	Scleral spur seen in all quadrants; patent ostium and PI superiorly
Fundus	Disc, retinal nerve fibre layer (RNFL); macula (hypotony maculopathy); choroidal detachment	Cup-disc ratio (CDR) 0.9, diffuse rim loss, laminar dot sign
Fields	Confrontation	"Fields (confrontation test): ____"

Special tests

Bleb — in slide order

1. Patient looks down: **height, clarity, horizontal extent** (clock hours), **location over the limbus**.
2. High magnification: **vascularity, microcysts, subconjunctival fibrosis, sutures**; then Seidel's test.

Record: in this order, ending "suggestive of a good functioning / failing / failed bleb".

Feature	Good functioning bleb	Failing or failed bleb
Height	Medium: low and diffuse, or circumscribed and elevated	Flat (reduced)
Clarity, wall	Translucent , thin	Opaque ; thick wall
Vascularity	Avascular	Highly vascular , corkscrew vessels
Microcysts	Numerous	Absent
Subconjunctival fibrosis	None	Present
IOP	Controlled	Raised

Seidel's test (bleb leak)

1. Fluorescein over the bleb: **0.25%** shows many leaks; **2%** shows subtle ones.
2. Cobalt blue light: aqueous leaking through the dye = **positive**.

Record: "Seidel's test negative."

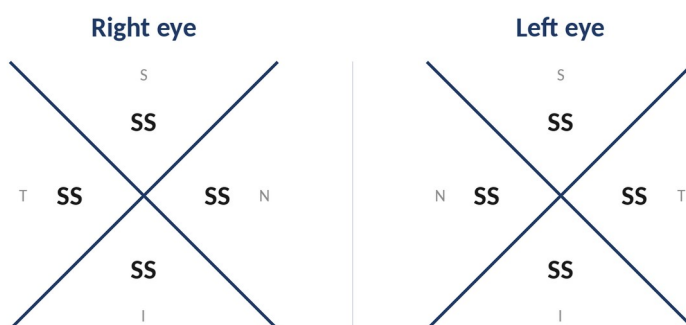
Retro-illumination (PI patency)

1. Light reflex clearly seen through the iridectomy = patent.
2. Margin **sharp, well defined** = **surgical**; **ragged, irregular**, with pigment dispersion = **laser**.

Record: "Retro-illumination positive: patent surgical PI at 12 o'clock."

Gonioscopy of the operated eye (technique: card G9)

1. After GAT, before dilatation: deepest structure seen in each quadrant.
2. Superiorly: internal ostium, PI, PAS.



Deepest structure seen in each quadrant — open angles in all quadrants (SS = scleral spur). RE: patent ostium and iridectomy superiorly

Summary and complete diagnosis

Mr ____, a ____-year-old man, had trabeculectomy (with ____) in the ____ eye ____ months ago; now on _____. Vision ____ / _____. Bleb ____ (slide order); Seidel's test _____. PI at ____ o'clock, patent. AC ____; lens _____. Tension by GAT ____ / ____ mm Hg at _____. Ostium _____. CDR ____ / _____. Fields ____.

eye → POAG → status post trabeculectomy (± antimetabolite) → bleb: functioning / failing / failed → IOP status → lens → other eye

- "Right eye: primary open-angle glaucoma status post trabeculectomy with failed bleb and grade 2 nuclear sclerosis. Left eye: primary open-angle glaucoma and grade 2 nuclear sclerosis."
- "Right eye primary open-angle glaucoma status post trabeculectomy with a good functioning bleb, intraocular pressure under control."

Differential diagnosis

Raised IOP after surgery:

Condition	For	Against
Failing bleb (salvageable)	Lower, thicker-walled, vascular; IOP rising	No microcysts, fibrosis obscuring details: no longer filters
Encapsulated bleb (type 4)	IOP raised; engorged vessels	Highly elevated dome; this bleb is flat
Blocked internal ostium	IOP raised; deep AC	Ostium patent on gonioscopy
Pupillary block, malignant glaucoma, delayed suprachoroidal haemorrhage	IOP raised	All flatten the AC; here AC normal, PI patent

What you must know

Definition

- **Trabeculectomy:** a **partial-thickness filtering operation**. Aqueous passes through a limbal fistula, around the edges of a scleral flap, into the subconjunctival space; it leaves by transconjunctival filtration or through lymphatics and vessels.
- **Failing bleb:** signs of failure appearing, IOP rising. **Failed bleb:** no longer filters (types 3 and 4), so IOP is elevated.
- For primary open-angle glaucoma itself, see card G1.

Classification

Filtering operations: full thickness, no flap (sclerectomy, trephination, Scheie's thermal sclerostomy, laser sclerostomy, iridencleisis, goniopuncture); partial thickness (**trabeculectomy**); non-penetrating (deep sclerectomy, viscocanalostomy, canaloplasty).

Bleb types (four)

Type	Appearance	Filters?
1	Thin, polycystic	Yes
2	Flat, thin, diffuse , relatively avascular; microcysts at high magnification	Yes
3	Flat , no microcysts, engorged surface vessels	No: IOP up
4	Encapsulated (Tenon's cyst): localised, highly elevated dome of hypertrophied Tenon's capsule; engorged vessels	No: IOP up

Pathogenesis

- Fibroblasts proliferate → **subconjunctival fibrosis**, the **commonest cause of bleb failure**; blood under the flap adds scarring.
- Long-term drops: more conjunctival fibroblasts and inflammatory cells, fewer goblet cells.
- Aqueous damps fibroblasts; older eyes do better (atrophic Tenon's capsule).

Investigations

Test	Why	Expected finding
Seidel's test	Bleb leak	Negative
Ultrasound biomicroscopy	PI patency; function of the trabeculectomy	Patent PI and ostium

Management

Indications (POAG): failed maximal medical therapy with progressive damage; IOP above target on 3 drops; poor compliance or side effects; baseline IOP over 40 mm Hg; one eye lost to glaucoma; difficult follow-up.

Relative contraindications: likely to respond to less invasive treatment; **previously failed trabeculectomy**; scarred conjunctiva; active neovascularisation; uveitic glaucoma.

Key steps

1. Bridle suture; limbus- or fornix-based conjunctival flap.
2. Scleral flap based at the limbus, **3 mm radially × 4 mm circumferentially**, cut through 2/3 of scleral thickness.
3. MMC now, if used (table).
4. Dissect the flap into clear cornea; paracentesis.
5. Enter the AC behind the flap hinge; Kelly Descemet's membrane punch: **internal ostium 1.5 × 2 mm**.
6. **Peripheral iridectomy**, basal, wider than the ostium.
7. Flap: interrupted **10-0 nylon**; reform the AC with balanced salt solution (BSS).
8. Conjunctiva: **8-0 vicryl**, watertight.
9. **Atropine 1%**; subconjunctival betamethasone and gentamicin.

Antimetabolites go in the sub-Tenon's space: fibrosis is the commonest cause of failure.

Feature	Mitomycin C	5-fluorouracil
Action	Alkylating agent; all phases of the cell cycle; much more potent	Pyrimidine analogue; S phase only
Dose	0.2–0.5 mg/ml for 1–5 min	50 mg/ml for 5 min ; subconjunctival 5 mg
Method	Sponge 4.5 × 4.5 mm under conjunctiva–Tenon's, flap edges pulled over it; rinse with BSS; tighter flap	30-gauge needle, 90°–180° from the site; twice daily week 1, daily week 2: 21 injections (105 mg)
Toxicity	More hypotony, bleb leak, bleb infection; endothelial ; cystic overfiltering bleb (commonest)	Epithelial ; punctal occlusion
Used in	Almost all; essential after failed trabeculectomy, conjunctival or cataract surgery, secondary glaucoma, the young or dark-skinned, drops over 3 years	MMC complications before; risk of hypotony maculopathy

Other modulation: steroids before and after surgery; non-steroidal anti-inflammatory drugs; collagen inhibitors (β-aminopropionitrile, penicillamine). Prevent toxicity: thorough irrigation, watertight closure.

Failing or failed filter

- **Blocked fistula** (early, deep AC): low-energy argon laser around the block; internal bleb revision; intracameral tissue plasminogen activator **6–12.5 µg** for fibrin or clot.
- **Failing bleb:** hourly steroids; **digital pressure** (IOP above **12 mm Hg** or target missed); argon laser suture lysis (**50 µm, 0.02–0.1 s, 250–1000 mW**) or releasable sutures.
- **Failed bleb** (late, fibrosis): restart drops; repeat filtering surgery with antimetabolite at a fresh site, or a drainage implant.
- **Encapsulated bleb:** settles in **2–4 months**; drops meanwhile.

This eye

1. Restart drops, **no beta-blocker** (asthma).
2. Above target: repeat trabeculectomy with **MMC** at a fresh superotemporal or superonasal site, or a **glaucoma drainage device** (card G3).
3. Cataract: temporal clear cornea spares the conjunctiva; combined surgery if IOP is uncontrolled.

Follow-up: bleb, leak, IOP, AC depth, disc and macula; fields as on card G1.

Complications

Early, besides the table: low IOP, formed AC → **hypotony maculopathy**; high IOP, deep AC → blocked fistula, failing bleb; also uveitis, hyphaema, dellen, loss of central vision.

Late: bleb infection and endophthalmitis, cataract, leaking bleb, Tenon's cyst, failure, malignant glaucoma, lid retraction, sympathetic ophthalmia. **Snuff-out:** sudden loss of vision after surgery in advanced glaucoma.

Shallow AC: commonest problem after surgery; causes PAS, endothelial damage, cataract.

Condition	IOP	Seidel	Fundus / other
Wound (conjunctival) leak	Low, soft eye	Positive	Poor bleb
Overfiltration	Very low	Negative	Good bleb ; choroidal detachment possible
Serous choroidal detachment	Low	—	Drain if very deep
Pupillary block (incomplete PI)	High	—	PI not patent
Malignant (ciliary block) glaucoma	High (hard eye) or normal	Negative	No bleb ; aqueous forced into the vitreous
Delayed suprachoroidal haemorrhage	High	—	Aphakic, highly myopic eyes at risk

— = not stated in the books.

- **Wound or bleb leak:** pressure dressing, Simmons shell, bandage contact lens, cyanoacrylate glue, aqueous suppressants; large defect → resuture or graft.
- **Overfiltration:** firm patching, shell, aqueous suppressants, atropine; then reform the AC.
- **Malignant glaucoma:** strong mydriatics → intravenous mannitol → neodymium:yttrium-aluminium-garnet (Nd:YAG) laser to the anterior hyaloid face through the PI → anterior vitrectomy via the pars plana.
- **Hypotony maculopathy:** prevent with a partial-thickness flap, tighter after MMC, and releasable sutures; 5-FU if at risk.
- **Bleb infection (blebitis):** earliest sign **mild hyperaemia around the bleb**; delay → **bleb-related endophthalmitis**. Frequent topical antibiotics; steroids after **12–24 hours**.

Prognosis and counselling

- Worse: age under 40, previous failed filter, aphakia or pseudophakia, neovascular glaucoma, active uveitis, previous conjunctival surgery.

- Plain trabeculectomy lowers IOP by about **41%**.
- Counsel: surgery protects the remaining vision; lost vision does not return. Come at once if the eye turns red.

Recent advances

- **Photodynamic therapy:** blue light with a photosensitising agent kills fibroblasts.
- **Non-penetrating surgery:** AC not entered, so less overfiltration and hypotony.

Viva questions

Why did you ask / examine...?

- Q Why did you ask when the operation was done?**
Early complications (shallow chamber, blocked fistula, failing bleb) differ from late ones (leak, infection, Tenon's cyst, cataract).
- Q Why did you ask whether an antimetabolite was used?**
More hypotony, bleb leak and bleb infection, especially with mitomycin C.
- Q Why did you ask about redness?**
Mild hyperaemia around the bleb is the earliest sign of bleb infection; delay risks endophthalmitis.
- Q Why did you ask about sudden loss of vision?**
Acute IOP rise with corneal oedema, central retinal vein occlusion, or snuff-out in advanced glaucoma.
- Q Why did you retro-illuminate the iridectomy?**
To prove patency: an incomplete iridectomy causes pupillary block, with high IOP and a flat chamber.
- Q Why did you do gonioscopy in an operated eye?**
To see the internal ostium and iridectomy, and any block: iris, ciliary body, lens, vitreous or clot.

Diagnosis

- Q What are the features of a good filtering bleb?**
Avascular; numerous microcysts; low and diffuse, or circumscribed and elevated.
- Q What are the signs of a failing bleb?**
Reduced height, thicker wall, vascularisation, loss of microcysts, raised IOP.
- Q What are the causes of failure of filtration?**
Intraocular: blocked fistula, bleb failure, poor technique. Extraocular: subconjunctival fibrosis (commonest), racial and genetic factors.

Investigations

- Q How do you detect a leaking bleb?**
Fluorescein: 0.25% shows many leaks, 2% subtle ones (Seidel positive).

Management and surgery

- Q Is "trabeculectomy" a misnomer?**
Yes. The block need not contain trabecular meshwork; it is more a peripheral posterior keratectomy.
- Q Why a peripheral iridectomy, and how big?**
To stop the iris blocking the ostium and to prevent pupillary block; basal and wider than the ostium.
- Q Why avoid 12 o'clock?**
The anterior ciliary vessel enters the ciliary body 3–4 mm from the limbus and bleeds; scarring there also blocks a repeat operation.
- Q Fornix-based or limbus-based conjunctival flap?**
Equal success. Fornix: easier, better exposure, early limbal leak. Limbus: suture lysis, 5-FU, digital pressure safer early; lower late IOP.
- Q When do you suture the scleral flap tighter?**
Angle-closure glaucoma or past malignant glaucoma; aphakic or highly myopic eyes; nanophthalmos; raised episcleral venous pressure.
- Q When do you advise digital pressure, and what are its risks?**
When IOP rises above 12 mm Hg or misses target. Risks: subconjunctival haemorrhage, hyphaema, wound rupture.
- Q Why use releasable sutures or laser suture lysis?**
The flap can be closed tight against hypotony, then IOP titrated after surgery.

Complications

- Q What is the commonest problem after trabeculectomy?**
A shallow anterior chamber.
- Q How do you tell wound leak, overfiltration and malignant glaucoma apart?**
Leak: soft eye, poor bleb, Seidel positive. Overfiltration: good bleb, Seidel negative. Malignant glaucoma: no bleb, Seidel negative, IOP high (hard eye) or normal.

Q Which organisms cause bleb-related endophthalmitis?

Streptococcus species (most common), Haemophilus influenzae, Staphylococcus.

Q What is the snuff-out phenomenon?

Sudden loss of vision after trabeculectomy in advanced glaucoma.

Recent advances

Q What is new in wound modulation?

Photodynamic therapy: blue light with a photosensitising agent kills fibroblasts.

Examiner traps

- A highly elevated bleb called “good”: an **encapsulated bleb** does not filter.
- “No leak” without **Seidel’s test**: leak and overfiltration both give a low IOP.
- Not retro-illuminating the **iridectomy**: an incomplete one causes pupillary block.
- Gonioscopy without naming the **internal ostium**.
- A **beta-blocker** for this asthmatic.
- “Status post trabeculectomy” without **bleb status, IOP and lens**.

Say-it script

SAY IT

My patient, a fifty-six-year-old asthmatic man, has gradual defective vision in the right eye for one year and the left eye for six months. He was not compliant with drops, and the right eye was operated six months ago.

Vision is 6/60 in the right eye and 6/24 in the left, improving to 6/12 and 6/9 with pinhole. The right eye has a flat, opaque bleb from eleven to two o’clock superiorly, with dilated tortuous vessels, no microcysts and subconjunctival fibrosis. Seidel’s test is negative. A surgical iridectomy at twelve o’clock is patent on retro-illumination. The anterior chambers are of normal depth. The right eye has a grade three relative afferent pupillary defect; both lenses have grade two nuclear sclerosis. Intraocular pressure by Goldmann applanation is twenty-six millimetres of mercury on the right and twenty-four on the left. Gonioscopy shows open angles, with a patent internal ostium on the right. The cup-disc ratio is 0.9 on the right and 0.75 on the left.

My diagnosis is right eye primary open-angle glaucoma status post trabeculectomy with failed bleb and grade two nuclear sclerosis; left eye primary open-angle glaucoma and grade two nuclear sclerosis.

I would document the fields, restart drops avoiding beta-blockers, and if the pressure stays above target, plan repeat trabeculectomy with mitomycin C at a fresh site or a glaucoma drainage device.

Quick recall

- **Good bleb**: avascular, numerous microcysts. **Failing**: lower, thicker-walled, vascular, microcysts lost.
- Commonest cause of failure: **subconjunctival fibrosis**; commonest problem after surgery: **shallow AC**.
- **Leak**: Seidel positive. **Overfiltration**: good bleb. **Malignant glaucoma**: no bleb; IOP high or normal.
- **MMC 0.2–0.5 mg/ml for 1–5 min; 5-FU 50 mg/ml for 5 min**, subconjunctival **5 mg**.
- Flap **3 × 4 mm**; ostium **1.5 × 2 mm**.
- Digital pressure if IOP above **12 mm Hg**; Tenon’s cyst settles in **2–4 months**.
- Earliest sign of bleb infection: **hyperaemia around the bleb**.

Mnemonic: none in your mnemonics file for this case.

Read more: Baidya p.151, 153–155, 164–166, 173, 177, 444 · Namrata p.172–173, 175–177 · Aravind 4.9, 4.18, 4.19, glaucoma model case sheet p.661–667

G3 POAG with a glaucoma drainage device

On the department checklist (not kept last year) · Long case

At a glance

- **Primary open-angle glaucoma (POAG)** with a **glaucoma drainage device (GDD)**: a tube in the anterior chamber (AC) drains aqueous to a plate at the equator, usually after a **failed trabeculectomy**.
- On the department checklist: stable patient, visible signs, long surgical viva.
- Examiners test the **tube**, **plate quadrant**, **motility**, intraocular pressure (IOP) control, device types, surgery, complications.

Case format — history

Mr X, a 62-year-old man, has come for regular follow-up; his right eye (RE) had a drainage device 3 years ago, after a failed trabeculectomy.

What to ask	Why it matters
History of present illness	
Trabeculectomy: how many, when, antimetabolite, result	Multiple failed trabeculectomies , failure with antimetabolites and scarred conjunctiva are the main indications
Device: eye, year, name (valved or not)	A non-valved tube is tied for 2–3 weeks ; IOP may rise before the tie goes
Early course: vision, admission, re-operation	Hypotony, flat AC, choroidal effusion, suprachoroidal haemorrhage
Double vision: passing or lasting	Passing: orbital and rectus oedema. Lasting: bleb displacing the globe, muscle fibrosis
Discomfort, redness, discharge	Dellen; patch graft melt, tube or plate erosion; endophthalmitis from tube exposure
Present drops: drug, strength, frequency	IOP falls less than after trabeculectomy; no miotics or alpha agonists
Negative history	
No history of (h/o) headache with vomiting, coloured haloes	Rules out an angle-closure attack
No h/o recurrent redness treated with steroid drops	Rules out uveitic glaucoma
No h/o trauma	Rules out post-traumatic (angle-recession) glaucoma
No h/o retinal laser or intravitreal injection	Rules out neovascular glaucoma
Past, drug and surgical	
Cataract surgery, penetrating keratoplasty (PKP)	Pseudophakic and post-PKP eyes may carry a pars plana tube ; graft failure risk
Diabetes, hypertension; asthma; sulphonamide allergy; warfarin	Neovascularisation, detachment risk; choice of drops (sulpha allergy bars carbonic anhydrase inhibitors); stop warfarin before surgery

Case format — examination

General and systemic (this case only)

- Pulse and chest: bradycardia or asthma bars beta-blockers.
- Face: a port-wine stain suggests Sturge-Weber syndrome, another glaucoma treated with devices.

Ocular examination — RE | LE template

Structure	Look for in this case	How to record (example)
Visual acuity (with pinhole)	Useful vision is a must; fall since surgery	6/18, pinhole (PH) 6/12
Ocular posture, extraocular movements (EOM)	Deviation, restriction, diplopia	Orthophoric; EOM full, no diplopia
Lids and adnexa	Upper lid over the plate; closure; blepharitis	Normal; complete closure
Conjunctiva (plate quadrant)	Bleb over the plate ; patch graft ; tube or plate exposure ; old trabeculectomy bleb	Bleb over plate superotemporally, no microcysts; graft intact; no exposure
Cornea	Tube–corneal touch ; oedema; graft clarity	Clear; no tube touch
Anterior chamber	Van Herick (VH) grade (flat in hypotony); cells, flare; hyphaema	VH grade IV (ND = normal depth); quiet

Structure	Look for in this case	How to record (example)
Tube	Clock hour; length; bevel; course; gap from cornea and iris; tip	10 o'clock, 2 mm, bevel up, parallel to iris, clear of cornea and iris; tip patent
Iris	Tube-iris touch ; peripheral iridectomy (PI); neovascularisation of the iris (NVI)	CPN (colour, pattern normal); PI at 12 o'clock, patent
Pupil, lens	As in master proforma; cataract progression	RTL (round, reacting to light) 3 mm; nuclear sclerosis grade 2
IOP	Goldmann applanation tonometry (GAT) before gonioscopy; time; drops	Tn (tension) by GAT 14 mm Hg at 10 am, 1 drug
Gonioscopy	Tube tip ; peripheral anterior synechiae (PAS); neovascularisation of the angle (NVA)	Cross diagram, deepest structure seen (below)
Fundus, fields	Disc (card G1); chorioretinal folds, hypotony maculopathy, choroidal effusion, detachment; then fields by confrontation	Cup-disc ratio (CDR) 0.8; no folds

Special tests

Slit-lamp examination of the tube

1. Diffuse light: entry site and clock hour.
2. Slit beam: length, bevel, course, gap from endothelium and iris, any touch.
3. Tip: open, or blocked by blood, fibrin, iris or vitreous.

Record: as in the table.

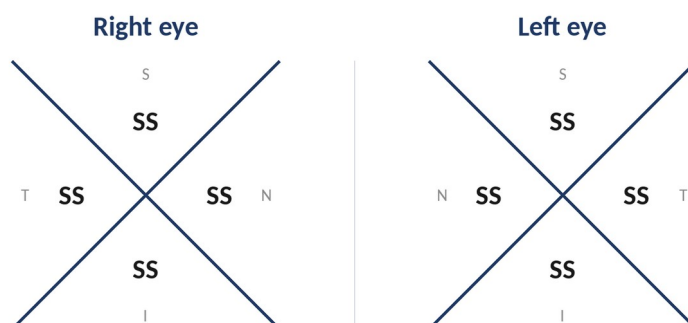
Plate quadrant and patch graft

1. Patient looks down and away from the plate; lift the upper lid.
2. Bleb over the plate in the house bleb order (height, clarity, extent, location, vascularity, microcysts, fibrosis, sutures); patch graft intact or melted; any exposure.

Record: as in the table.

Gonioscopy with a tube

1. IOP first: gonioscopy lowers the next GAT reading.
2. One cross for the RE and one for the left eye (LE), deepest structure per quadrant; note the tube tip, PAS, NVA.



Deepest structure seen in each quadrant (SS = scleral spur) — open angles in all quadrants. RE: tube tip free superotemporally; no PAS or NVA

Summary and complete diagnosis

- Mr ___, a ___-year-old man with POAG for ___ years, had a failed trabeculectomy and a drainage device in the ___ eye; he uses ___. That eye: tube at ___ o'clock, ___ mm, bevel ___, tip ___; bleb ___; graft ___; EOM ___; IOP ___ mm Hg (GAT, ___ am); gonioscopy ___; CDR ___. Other eye: ___.
- **Eye → POAG → stage → status: operations → tube → plate → IOP control → lens → other eye**
- "Right eye primary open-angle glaucoma, severe stage, status post trabeculectomy with failed bleb and glaucoma drainage device implantation, with a well-positioned patent tube, intraocular pressure under control on medical management, and grade 2 nuclear sclerosis; left eye primary open-angle glaucoma, moderate stage, on medical management."
- "Right eye primary open-angle glaucoma status post glaucoma drainage device implantation with a patent tube and intraocular pressure above target, suggestive of a thick fibrous capsule."

Differential diagnosis

Refractory secondary glaucomas that are also treated with devices:

Condition	For	Against
Neovascular glaucoma	Device often the first operation	No NVI or NVA; no retinal laser

Condition	For	Against
Uveitic glaucoma	Indication; chronic iritis after a device	No recurrent red eye; quiet AC
Angle-recession glaucoma	Unilateral; post-traumatic indication	No trauma; no widened ciliary body band
Glaucoma after PKP	Indication; pars plana tube indicated	No graft

What you must know

Definition

- **Glaucoma drainage device (aqueous shunt)**: an **alloplastic tube** leading to a **large equatorial reservoir** that drains aqueous humour from the AC.
- **Parts**: **tube** (AC or pars plana); **plate (explant)** at the equator, with a **ridge** where the tube inserts (keeps the capsule off the tube opening); **valve** if valved; after surgery, a **capsule (bleb)** and a **scleral patch graft**.
- All devices: materials fibroblasts cannot firmly adhere to; equatorial plate; similar tube diameters. For POAG itself, see card G1.

Classification

By flow: **seton** (non-lumened; flow along its surface) · **shunt** (hollow; two-way flow) · **valve** (one-way; needs an activating pressure).

Named devices: valved vs non-valved

Device	Valve	Plate	Tube
Molteno (the first)	Non-valved	Circular, polypropylene; single, double or triple	Outer 0.63 mm , inner 0.3 mm
Baerveldt	Non-valved	Silicone	Outer 0.64 mm , inner 0.3 mm
Ahmed glaucoma valve	Valved	Oval polypropylene, 184 mm² (double 364 mm²)	Silicone; opens above 8–10 mm Hg

- Ahmed: two silicone elastomer membranes, **Venturi effect** (Bernoulli's principle): fully open at high pressure, narrowing as pressure falls.
- Non-valved sizing: **larger plates for younger, healthier eyes**; smaller for older, sicker eyes.

Pathogenesis

- Aqueous runs down the tube onto the plate; a **fibrous capsule** forms around the plate and becomes the bleb.
- **Drainage capacity** $\propto$ **capsule surface area**; **resistance** $\propto$ **capsule thickness**.
- Valved or flow-restricted devices work at once; a non-valved tube waits for the capsule.

Investigations

Test	Why	Expected finding
GAT: time, drops	Low = hypotony; high = blocked tube, thick capsule, hypertensive phase	At target
Gonioscopy	Tube tip; PAS (avoid for a tube); NVA (treat first)	Tip free; open angle
+90 dioptre (D) lens; indirect ophthalmoscopy	Disc; folds, choroidal effusion, detachment	No hypotony signs

Corneal oedema: take the IOP as higher than read; repeat GAT once it clears.

Management

This case (device working): drops to target as for POAG (card G1), but **no miotics or alpha agonists**: they increase vascularisation of the bleb capsule. Every visit: tube, cornea, plate cover, motility, IOP.

Indications: refractory glaucomas (uveitic, neovascular, congenital, refractory juvenile, after PKP, aniridia, post-traumatic); **multiple failed trabeculectomies** or failed filtering surgery with antimetabolites; **extensive conjunctival scarring**; failed primary glaucomas.

Before surgery (contraindications and checks)

- **Potential for useful vision is a must**; without it, cyclodestruction.
- Check lid position, scleral exposure, tear film, blepharitis, peripheral cornea, vitreous in the AC (may block the tip), cataract (combine), PAS (avoid for the tube), NVA (treat first).
- Pseudophakia, shallow AC, post-PKP, poor endothelium → **pars plana tube**. Stop **pilocarpine** or **echothiophate** and **warfarin**.

Key steps

1. Limbus-based or fornix-based conjunctival flap.
2. **Superotemporal** quadrant (next, inferonasal); avoid superonasal.

3. Plate at the **equator** between two recti, anterior edge **8–10 mm** behind the limbus, sutured to **episclera** (superior rectus insertion if staphyloma or quiescent scleritis).
4. **Non-valved: tie the tube for 2–3 weeks (rip cord** of 4-0 chromic catgut; **8-0 polyglactin or 10-0 prolene** released by **argon laser suturolysis**; 5-0 nylon with a spacer). **Valved: one stage.**
5. **AC tube: 23-gauge needle tract** at the posterior edge of the blue limbus; **bevel up**; **1.5–2 mm** inside, **parallel to the iris.**
6. Or **pars plana tube**: complete vitrectomy; **3–4 mm** behind the limbus; **bevel down.**
7. Full-thickness **scleral patch graft** over the **3–5 mm** of tube exposed to lid contact.
8. Reform the AC with **hyaluronic acid (Healon)**: less hypotony and flat AC.

After surgery: a **hypotensive phase** and a **hypertensive phase** (IOP poorly controlled although the device works properly).

Valved vs non-valved: valved is one-step; safety and efficacy appear equal, except in Sturge–Weber syndrome, where the **Ahmed valve** gives better long-term control than the Molteno.

Device vs trabeculectomy

Feature	Drainage device	Trabeculectomy
Bleb	Posterior ; organised capsule; no microcysts	At the limbus; microcysts
IOP fall	Smaller percentage ; comparable results in neovascular glaucoma	Larger
Place in POAG	After failed filtering surgery or scarred conjunctiva	First operation after failed medical treatment

Still uncontrolled, or poor visual potential: cyclodestruction (transscleral diode or endolaser cyclophotocoagulation) reduces aqueous production; risks hypotony, phthisis bulbi, retinal detachment; may need repeating.

Complications

Intraoperative: hyphaema; lens or endothelial damage; **scleral perforation** while suturing the plate (gush of vitreous → **cryotherapy** under indirect ophthalmoscopy); hypotony.

Complication	Cause	Prevention / action
Early: hypotony, flat AC, choroidal effusion, suprachoroidal haemorrhage	Excess run-off; leak around the tube ; high pre-operative IOP	Tight tract; tie; Healon
Early: IOP rise before the tie absorbs	Tied tube	Danger to an advanced disc
Tube occlusion	Blood, fibrin, iris, vitreous	Full vitrectomy for a pars plana tube
Tube migration: forward onto cornea, lens or iris; or extrusion	Hypotony; poor plate fixation	Correct length, parallel to iris; secure plate
Graft melt, erosion → endophthalmitis	Poor lid closure, dry eye; exposure	Patch graft; check lids
Raised IOP, open tube	Thick fibrous capsule ; hypertensive phase	No miotics or alpha agonists
Diplopia, strabismus	Bleb displacing globe; fibrosis; superonasal plate (pseudo-Brown's syndrome)	Superotemporal plate

Also: corneal oedema and graft failure (with or without touch); dellen; aqueous misdirection; hypotony maculopathy, chorioretinal folds; cataract; chronic iritis; retinal detachment; early endophthalmitis (contamination at surgery).

Prognosis and counselling

- IOP falls less than after trabeculectomy; drops may still be needed.
- Report at once: pain, redness, discharge, falling vision (infection through an exposed tube) or double vision.

Recent advances

- Neovascular glaucoma with vitreous haemorrhage: intravitreal anti-vascular endothelial growth factor (anti-VEGF) **2–3 days** before vitrectomy, panretinal photocoagulation (PRP) and **valve implantation** gives a high success rate.

Viva questions

Why did you ask / examine...?

Q Why did you ask about previous trabeculectomy?

Multiple failed trabeculectomies and scarred conjunctiva are the main indications.

Q Why did you ask about double vision?

Early, transient: orbital and rectus oedema. Late: bleb displacing the globe, muscle fibrosis.

Q Why did you ask which drops he uses?

Miotics and alpha agonists are contraindicated: they increase vascularisation of the bleb capsule.

Q Why did you lift the upper lid?

To see the bleb, patch graft and any exposure; tube exposure is linked to endophthalmitis.

Q Why did you look at the gap between tube, cornea and iris?

Tube–corneal touch causes corneal oedema and graft failure; in hypotony the tube moves forward.

Diagnosis

Q Define a glaucoma drainage device.

An alloplastic tube leading to a large equatorial reservoir that drains aqueous humour from the AC.

Q What are setons, shunts and valves?

Seton: non-lumened, surface flow. Shunt: hollow, two-way flow. Valve: one-way, needs an activating pressure.

Q How does a device bleb differ from a trabeculectomy bleb?

More posterior; organised capsule separable from Tenon's capsule; no conjunctival microcysts.

Investigations

Q What do you look for on gonioscopy in a device eye?

Tube tip (free or blocked); PAS (keep a tube away); NVA (treat before surgery).

Management and surgery

Q How does the Ahmed valve work?

Two silicone elastomer membranes act as a one-way valve (Venturi effect), opening only above 8–10 mm Hg.

Q Where do you place the plate, and why not superonasally?

Superotemporally, at the equator between two recti, 8–10 mm behind the limbus. Superonasal: pseudo-Brown's syndrome, optic nerve impact.

Q Why and how is a non-valved tube tied?

For 2–3 weeks while the capsule forms: rip cord, a laser-released suture, or nylon with a spacer.

Q When do you choose a pars plana tube?

Aphakia or pseudophakia, after PKP, shallow AC, poor endothelium.

Q Why a scleral patch graft, and its complications?

It covers the 3–5 mm of tube exposed to lid contact, preventing tube erosion, hypotony, tube–corneal touch and scleral perforation; complications: dellen, scleral melt.

Q Which drugs do you stop before surgery?

Pilocarpine or echothiophate (less postoperative inflammation) and warfarin.

Complications

Q What causes hypotony after one-stage and two-stage surgery?

Leak around the tube where it passes through the eye wall; a needle tract gives a tight fit.

Q How do you recognise and treat scleral perforation?

A gush of vitreous; cryotherapy to the area under indirect ophthalmoscopy.

Q What are the phases after device surgery?

Hypotensive and hypertensive; in the hypertensive phase IOP control is inadequate although the device works.

Q What if the pressure stays high despite surgery?

Cyclodestruction (diode or endolaser cyclophotocoagulation); risks hypotony, phthisis bulbi, retinal detachment.

Recent advances

Q How do you improve valve surgery in neovascular glaucoma?

Intravitreal anti-VEGF 2–3 days before vitrectomy with PRP and valve implantation.

Examiner traps

- Calling the bleb over the plate a trabeculectomy bleb, or expecting microcysts.
- "Tube in the AC" without clock hour, length, bevel, course, gaps and tip.
- Not lifting the upper lid to see the plate, graft and exposure.
- Skipping EOM and diplopia (on the department sheet).
- Advising pilocarpine or an alpha agonist after a device.

Say-it script

SAY IT

Mr X, a sixty-two-year-old man with primary open-angle glaucoma in both eyes for ten years, has come for regular follow-up. His right eye had a failed trabeculectomy and then a glaucoma drainage device three years ago. He has no double vision or redness.

In the right eye, a tube enters the anterior chamber at ten o'clock: two millimetres long, bevel up, parallel to the iris, clear of the cornea and iris, with a patent tip. A bleb lies over the plate superotemporally; the patch graft is intact, with no exposure. The chamber is quiet and of normal depth, and the lens shows grade two nuclear sclerosis. Eye movements are full, without diplopia. Intraocular pressure by Goldmann applanation tonometry is fourteen millimetres of mercury on the right and sixteen on the left at ten in the morning. Gonioscopy shows open angles with the tube tip free. The cup-disc ratio is zero point eight on the right and zero point six on the left.

My diagnosis is: right eye, primary open-angle glaucoma, severe stage, status post trabeculectomy with failed bleb and glaucoma drainage device implantation, with a well-positioned patent tube, intraocular pressure under control on medical management, and grade two nuclear sclerosis; left eye, primary open-angle glaucoma, moderate stage, on medical management.

I would continue his drops, avoiding miotics and alpha agonists, review the tube, cornea, plate and pressure at every visit, and repeat the fields.

Quick recall

- Drainage $\propto$ capsule area; resistance $\propto$ capsule thickness.
- **Ahmed**: valved, **184 mm²**, opens above **8–10 mm Hg**; **Molteno**, **Baerveldt**: non-valved, **tied 2–3 weeks**.
- Plate **superotemporal**, **8–10 mm** behind the limbus; avoid superonasal (pseudo-Brown's).
- AC tube **bevel up**, **1.5–2 mm**, parallel to iris; pars plana **bevel down**, **3–4 mm** behind limbus.
- Device bleb: posterior, **no microcysts**; smaller IOP fall than trabeculectomy.
- After a device: **no miotics**, **no alpha agonists**.
- Hypotony after either technique: **leak around the tube**.

Mnemonic: none in your mnemonics file for this case.

Read more: Baidya p.165, 173–177, 179–180 • Namrata p.175–177, 186, 195, 198 • Aravind 4.20, glaucoma model case sheet p.661–663